



VCS JOINT VALIDATION & VERIFICATION REPORT FOR GROUPED HYDRO POWER PROJECT BY EKI ENERGY SERVICES LIMITED



Document Prepared by TÜV SÜD South Asia Pvt Ltd

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Summary:

TÜV SÜD has performed joint validation and first verification for current monitoring period from 17-September-2021 to 31-December-2022 of grouped project titled as GROUPED HYDRO POWER PROJECT BY EKI ENERGY SERVICES LIMITED ('the Project') which is listed under VCS registry with project ID 4139 started on 17-September-2021.

TÜV SÜD has performed joint validation and first verification as per the requirements of section 4 of VCS Standard Version 4.5 following the guidance of Validation and Verification Manual. Validation & Verification Team has used risk-based approach to provide conformance with ISO 14064-3 and ISO 14065. Validation & Verification Team has conducted independent assessment of estimated and actual GHG statement of emission reductions and removals that have occurred as a result of the project implementation during this monitoring period to be materially correct and conform claim against the requirements of VCS Program rules and applied methodology.

Validation & Verification Team has validated and verified that the statement of historical data and information of the project to a reasonable level of assurance and ensures that the project meets the relevant materiality requirements (misrepresentation<5% for small scale project) through means of onsite assessment as per the audit criteria. Validation & Verification Team has not identified any uncertainties which might impact GHG statement of emission reductions and removals as result of risk-based approach.

During joint validation and first verification for the current monitoring period from 17-September-2021 to 31-December-2022 of grouped project titled as GROUPED HYDRO POWER PROJECT

BY EKI ENERGY SERVICES LIMITED ('the Project') which is listed under VCS registry with project ID 4139, 13 Clarification Requests (CL), 20 Corrective Action Requests (CARs) and 00 Forward Action Requests (FARs) are raised, addressed and closed out by VVB.

Based upon the conclusion of objective evaluation process against the criteria of applicable methodology and project standard under the scope of validation and verification Herby, TÜV SÜD concludes that GROUPED HYDRO POWER PROJECT BY EKI ENERGY SERVICES LIMITED ('the Project') which is listed under VCS registry with project ID 4139, has achieved the actual GHG Emission Reductions 26,567 tCO₂e through electricity generation during the current monitoring period from 17-September-2021 to 31-December-2022 whereas annually estimated GHG Emission Reductions are 21,741 tCO₂e.

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1 INTRODUCTION

1.1 Objective

TÜV SÜD has been commissioned by the client to perform an independent joint validation first verification assessment. The objective of the verification work is to comply with the requirements of Verified Carbon Standards requirements. Validation & Verification Team's objective is to ensure that:

- Validation of project activity to conform eligibility against the requirements of VCS program.
- Evaluation of the reasonableness of assumptions, limitations, and methods in order to assess and determine whether the project and its GHG statement conforms with the VCS Program rules for future activities.
- Assessment of appropriateness of applied methodology, methodological tools and applicable methodological regulatory in accordance with the VCS program rules and requirements.
- Project activity has been implemented and operated as per description in PD-MR, and that all physical features (technology, project equipment, monitoring and metering equipment) of the project are in place.
- Published PD-MR and associated objective evidence provided are sufficient, verifiable and in accordance with applicable VCS rules and requirements.
- Actual monitoring systems and procedures comply with the monitoring systems and procedures described in the registered monitoring plan and the approved methodology.
- Objective Information is recorded and preserved as per the applicable requirements.
- Assessment of the sustainability monitoring parameters as per the VCS Program rules.

1.2 Scope and Criteria

The scope of assessment is defined by the underlying legislation, regulation and guidance given by relevant entities or authorities. In the case of VCS project activities, the scope is set by:

- VCS standard version 4.5 requirements.
- Baselines and monitoring methodologies (including GHG inventories).
- Environmental issues relevant to the applicable sectoral scope.

- Current technical and operational knowledge of the specific sectoral scope and information on best practice.
- Stakeholder consultation and feedback

The joint validation and first verification process is not meant to provide any form of consulting for the project participant (PP). However, stated requests for clarifications, corrective actions, and/or forward actions may provide input for improvement of the project design.

1.3 Reasonableness of Assumptions and Level of Assurance

The assumptions, limitations, and methods to support any statements are assessed against the requirements of VCS program guide, VCS Standard version 4.5 and applied methodology/methodological tools. The errors identified in the project are within threshold limit of materiality as per the conclusion of assessment under scope of joint validation and verification hence, GHG emission reductions are calculated without material misstatements. The VVB confirms that a reasonable level of assurance has been achieved during the validation and verification process.

1.4 Summary Description of the Project

This project activity is identified as grouped activity which involves electricity generation via utilizing the hydropower (run of river) through installation of 2 turbines in Ilam District, Province 1 of Nepal, one is having capacity of 9.51 MW Mai Beni Hydropower Project (MBHP) and other 6.2 MW Lower Jogmai Khola Hydropower Project (LJKHP) thus the overall capacity of this project activity is 15.71 MW.

Mai Beni Hydropower Project (MBHP) is located within Mai-Jogmai Rural Municipality and Ilam Municipality, Ilam District, State-1 in the Eastern Region of Nepal. The project site is about 527 km east of Kathmandu. The headworks site is located near the border of Soyang and Namsaling. The powerhouse is located 900m downstream of Mai khola and Jogmai khola confluence.

Whereas, Lower Jogmai Khola Hydropower Project (LJKHP) is located at Jogami River of Ilam District, Mechi Zone of Eastern Development Region, Nepal. Project site is accessible by Ilam-Namsaling motorable road from Tilkenimode to Powerhouse Site which is about 6 km. The same road is reaching to surge tank which is about 7.25 km from Tilkenimode. The powerhouse can also be reached by motorable road about 3.5 km from Silpur which is opposite bank of the powerhouse.

The proposed activity intends to provide clean energy in sustainable manner by reducing GHG emissions anthropogenic emissions of greenhouse gases into the atmosphere which would have been generated otherwise via common practice of fossil fuel-based power generation in absence of this project activity. Over a first crediting period project activity is supposed to achieve annual estimated emission reductions of 21,741 tCO₂e.

The Project has received necessary approvals for development and commissioning for each plant from the state Nodal agencies and is in compliance to the local laws and regulations. Project proponent has provided supportive evidences which assessed by VVB team to confirm that the project is complying following laws. List of Project Related Policies, Acts and Regulations are as follows -

Legislations	Functions
Company Act 2053 (1996) ¹	Registration and operation
Hydropower Development Policy 2058 (2001) ²	Guidelines for hydropower development
National Biodiversity Strategy and Action Plan 2071 B.S. (2014) ³	Management of Diversity
Electricity Act 2049 (1992) ⁴	Regulations for survey, generation, transmission and distribution of Small Hydropower power
Water Resources Act 2049 (1992) ⁵	Regulations for rational utilization, conservation,
NEA's policy regarding hydropower 2055 (1998) ⁶	Regulation regarding buy back rates and Power purchase agreement of small hydropower projects by private sectors
Environment Protection Act 2054 (1997) ⁷ Environment Protection Regulations (as amended) 2055(1999)	Regulations regarding environment protection and hydropower development

¹ https://nepalindata.com/media/resources/bulkuploaded/companies_act_2053_eng_april_17.pdf

² <https://www.lawcommission.gov.np/en/wp-content/uploads/2018/09/hydropower-development-policy-2058-2001.pdf>

³ <https://www.cbd.int/doc/world/np/np-nbsap-v2-en.pdf>

⁴ <https://admin.theiguides.org/Media/Documents/ElectricityAct1992.pdf>

⁵ <https://admin.theiguides.org/Media/Documents/WaterResourcesAct1992.pdf>

⁶ https://www.nea.org.np/admin/assets/uploads/supportive_docs/1998-99-compressed.pdf

⁷ <https://faolex.fao.org/docs/pdf/nep100343.pdf>

National Energy Crises Policy, 2073⁸

To improve Energy Crisis in Nepal

2 VALIDATION AND VERIFICATION PROCESS

2.1 Method and Criteria

The information provided by the project participants is assessed by applying the means of validation and verification to conform compliance with specified in the VCS Standard Version 4.5 following ISO 14064-3 and ISO 14065 using risk-based process.

A competent Validation & Verification team is selected prior to the start of the validation and verification. The team is selected to cover the technical area(s), sectoral scope(s) and relevant host country experience for evaluating the VCS project activity. Additionally, a competent Technical Reviewer or Technical Reviewer Team is appointed to conduct checks on quality and completeness.

Table 2.1.1 Composition of Team under scope of Validation and Verification

Role/Qualification	Last Name	First Name
Team Leader/Validator & Verifier	Kudtarkar	Shruti
Validator & Verifier	Kudtarkar	Shruti
Technical Expert	Shinde	Milind
Trainee Validator & Verifier	Chauhan	Ankita
Technical Reviewer	K	Sudheendra

The Validation & Verification team performs evidence gathering and desk review to evaluate the assumptions, limitations, and methods that support a statement in reasonable manner about the outcome of future activities in order to determine conformance with VCS program and rules also, to verify historical data and information meets reasonable level of assurance to assess relevant materiality requirements and fulfillment by the project activity.

Following direct communication followed by on-site visit of facilities and project area and verification of project related objective evidence with respect to project Implementation and

⁸ [https://aepc.gov.np/uploads/docs/2018-06-19_RE%20Subsidy%20Policy,%202073%20\(English\).pdf](https://aepc.gov.np/uploads/docs/2018-06-19_RE%20Subsidy%20Policy,%202073%20(English).pdf)

operation to conform compliance of project activity against the requirement of VCS program and applied methodology via assessing aggregation, representation and reporting of GHG emission reductions and/or removals and associated qualitative information which results in assessment conclusion and formation of a draft joint validation and report. This joint validation and report and other supporting documents then undergo an internal quality control by the Certification Body of “Environment and Energy” before submission to the VCS. In case any deviation is proposed or identified Validation & Verification team evaluates the request for deviation as per guidance in VCS program guide version 4.3 and VCS Standard Version 4.5.

2.2 Document Review

Desk review is carried out by Validation & Verification Team to confirm sufficiency of evidence and associated data and information to be verified for project development stage, project implementation phase and current monitoring period through completeness check (i.e., identification of source and nature of objective evidence). Validation & Verification Team reviewed project description, details of project proponents, geospatial information of project i.e., site numbers and locations, development and implementation of project activity, development of monitoring plan and implementation, applied methodologies, applied regulatory documents, measurement frequency of monitored parameters, calibration frequency of monitoring equipment. Also, reviewed and evaluated quality assurance and quality control procedures with respect to their influence on generation and reporting of GHG emission reductions or net anthropogenic GHG removals.

Documents reviewed to verify objective evidence for joint validation and first verification:

- Project Design-Monitoring Report (PD-MR).
- Associated Emission Reductions calculation data and information, criteria established in monitoring report against requirements of Applied methodologies other applied methodological regulatory documents.
- Proof of Commissioning.
- Calibration records.
- System, technology and equipment installed at project site.
- Logbook and records of monitoring and measurement, supply/consumption related invoices
- Quality assurance and quality control procedures.

Validation & Verification Team crosschecked PD-MR information with respect to objective evidence to determine whether the provided quantitative are accurate, refer to List of review document in Appendix [2] for the same.

2.3 Interviews

Validation & Verification Team has conducted interviews during site visit to discuss with respective personnel to verify the data and records pertaining to project activity under scope of validation and verification in successfully manner.

Table 2.3.1 Summary of conduct of on-site interviews under scope of validation and verification

S.N.	Interviewee			Date	Subject	Team member
	Last name	First name	Affiliation			
1.	Mandal	Santosh	Deputy Plant Manager at Mai Beni Hydropower Project (MBHP) and Lower Jogmai Khola Hydropower Project (LJKHP)	03/07/2023, 04/07/2023	Project Implementation , Operations, Monitoring & Measurement	Shruti Kudtarkar and Ankita Chauhan

2.4 Site Visits

Validation & Verification Team has performed on-site assessment on 03/07/2023 and 04/07/2023 fulfilling all the criteria of section 4 of VCS Standard Version 4.5 as per guidance

of Validation and Verification Manual following stepwise standard auditing techniques as below:

- Assessed Implementation and operation of GROUPED HYDRO POWER PROJECT BY EKI ENERGY SERVICES LIMITED ('the Project') in accordance with the PD-MR and confirmed during on-site assessment.

Project activities are located at Mai-Jogmai Rural Municipality and Illam Municipality, Illam District, State-1 in the Eastern Region of Nepal (MBHP) and Jogami River of Ilam District, Mechi Zone of Eastern Development Region, Nepal (LJKHP).

Table 2.4.1 Summary of physical location for on-site assessment of project facilities

Project Identity	Installation Capacity (MW)	Physical Location
MBHP	9.51	Mai-Jogmai Rural Municipality and Illam Municipality, Illam District, State-1, Eastern Region, Nepal
LJKHP	6.2	Jogami River of Ilam District, Mechi Zone, Eastern Development Region, Nepal

- Validated the project implementation against the reporting in PD-MR.
- Cross-checked the ownership and facility details against the reporting in PD-MR.
- Validated the technology, system, equipment specifications against the reporting in PD-MR.
- Reviewed information flows for generating, aggregating and reporting monitoring parameters and confirmed during on-site assessment.
- Interviewed with relevant personnel to determine whether operational and data collection procedures are implemented in accordance with registered monitoring plan/validated permanent changes to monitoring plan during on-site assessment.
- Cross checked between information provided in monitoring report and data from consolidated in ER (Emission Reductions) sheets sourced from plant logbooks, inventories, purchase records and confirmed during on-site assessment.
- Checked monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the PD-MR, the applied methodologies and the other applied methodological regulatory documents which is to be confirmed during on-site assessment.
- Reviewed calculations and assumptions made in determining the GHG data and GHG emission reductions or net anthropogenic GHG removals.
- Identified quality control and quality assurance procedures in place to prevent, or identify and correct, any errors or omissions in the reported monitoring parameters.

2.5 Resolution of Findings

Validation & Verification Team raised on 05-July-2023 and 13-July-2023 15 Clarification Requests (CL), 29 Corrective Action Requests (CARs) and 00 Forward Action Requests

(FARs) under scope of joint validation and verification of grouped project titled as GROUPED HYDRO POWER PROJECT BY EKI ENERGY SERVICES LIMITED ('the Project') which is listed under VCS registry with project ID 4139. Validation & Verification Team has followed the VCS standard requirements to evaluate the PP's response to respective individual CL, CAR and FAR and successful closing of the same. Refer to Appendix [1] of details on description, PP's response and conclusion of VVB's assessment for each CL/CARs/FAR.

2.5.1 Forward Action Requests

As it is the Joint validation and first verification there are no previous FARs to be identified by Validation & Verification Team.

3 VALIDATION FINDINGS

3.1 Project Details

Validation & Verification Team has identified this project activity is as grouped activity which involves electricity generation via utilizing the hydropower (run of river) through installation of 2 turbines in Illam District, Province 1 of Nepal, one is having capacity of 9.51 MW Mai Beni Hydropower Project (MBHP) and other 6.2 MW Lower Jogmai Khola Hydropower Project (LJKHP) thus the overall capacity of this project activity is 15.71 MW. Where, electricity generated is supplied to Integrated Nepal Power System (INPS) through 4 km long, 33 kV Single Circuit Transmission line from the project to the 132/33kV Godak sub-station of NEA. As per the submitted PD-MR the PPs are Urja Developers Private Limited and EKI Energy Services Limited (whereas EKI Energy Services Limited is holding the right for communication on behalf of registry user as per "Verra-Registry-Communications-Agreement-multiple-PPs.").

Validation & Verification has assessed and verified the technology and measures implemented by Samling Power Company Ltd. And Asian Hydropower Ltd. For the grouped project and same are cross checked with the objective evidence through desk review and on-site assessment.

Validation & Verification Team has assessed the eligibility of the project under scope of VCS program against the requirement of section 2 of VCS standard and same is found in line.

Validation & Verification Team has identified and adequacy of response against the eligibility criteria of this grouped project VCS standard version 4.5 and applied methodology conclusion upon the same is stated below as per the response provided:

Table 3.3.1 Assessment of eligibility criteria for grouped project

Sr.No	Eligibility Criteria	Project Activity instances eligibility	VVB Assessment
1	Applicability Conditions: The project activity instances shall meet applicability conditions for applicable methodology (AMS I.D version 18.0) as defined in section 3.2.	All project activity instances meet applicability conditions of methodology (AMS I.D Version 18.0 (project activity instances having capacity less than 15 MW), hence this eligibility criteria are fulfilled. Please refer section 3.2 of this document for methodology applicability criteria.	All the instances involved are capacity less than 15 MW thus fulfill the requirement.
2	Geographical Area: The project activity instances to be included in the grouped project activity will be activities involving Renewable Electricity Generation and grid using hydropower plant located within Nepal.	All project activity instances being included in the grouped project are located within geographic boundaries of Nepal. Hence this condition is fulfilled. Please refer commissioning certificate for Geographical area of project activity.	All the instances involved are within geographical boundaries of Nepal.
3	Baseline scenario: : All Project Activity Instances shall meet the baseline definition as defined in respective methodology and as explained in section 3.4.	All initial project activity instances have same baseline as per methodology as detailed in subsequent sections. Hence this eligibility criterion is fulfilled. Please refer section 3.4 of this document for baseline scenario.	All instances have same baseline i.e. grid electricity. Thus fulfil the requirement.
4	Technology type: The project activity instances to be included in the grouped project activity will be Greenfield activities involving Renewable Electricity Generation and grid using hydropower. The electricity should be supplied to grid.	All project activity instances are hydro power plants supplying electricity for grid. All are green field projects. Hence this condition is fulfilled	All the instances involved are of same technology i.e. Greenfield Hydro power and supplying electricity for grid only, thus fulfil the requirement, thus same has been verified through power purchase agreement
5	Additionality: The project activity instances to be added as part of the grouped project will meet additionality criteria as set out in respective methodologies - for Small Scale Project instances the methodology applicable will be AMS-I.D.	The additionality for the all-project activity instances included in the grouped project has been demonstrated in section 3.5 of PD.	All instances are demonstrated additionality with investment analysis and same are in line with TOOL21- Demonstration of additionality of small-scale project activities, Version 13.1. This has been further confirmed

			by VVB team through IRR sheet submitted by project activity for both instances.
6.	Start Date: The start date of each project activity instance under the grouped project should not be prior to the start date of the grouped project. The start date of each project activity instance will be determined through documentary evidence.	The start date of grouped project is considered as the earliest commissioning date of project activity instance and start date of all other project activity instances is after this date. Please refer commissioning certificate for start date of grouped project activity.	Start date of all instances are after 17-September-2021 thus fulfill the requirement
7	Conditions that avoid double counting of emission reductions like unique identifications of project and claiming emission reduction only under one GHG program	The existing and new project activity instances are identified by latitude and longitude and location of the plant. Declaration has been provided and these project activity instances are not participating in any other GHG program. Hence this condition is fulfilled	Unique locations of the instances are provided in the Joint VCS PD & MR and PP have also submitted declaration for avoidance of double counting of the emission reductions achieved. Thus, fulfill the requirement.

Additionally, as per clause 3.6.89 of VCS Standard Version 4.5, the distance between two instances is more than 10 KM, thus fulfils the requirement.

Further, each project activity instances are not exceeding one percent of the capacity limit shall be identified of 15 MW. The project activity instance involves multiple project activity instances and it is not cluster of project activity instances. Also, each project activity instances not exceeding the 1% of capacity limit of 15 MW. Thus, there is no any requirement of division of clusters of project activity instances and no requirement to demonstrate that each project activity instances are not within one kilometre of each other.

Furthermore, as per clause 3.6.127 of VCS Standard Version 4.5, Grouped project activity allows multiple project activities where the methodology applied to project allows it. "As with non-grouped projects, grouped projects may incorporate multiple project activities (see Section 3.6.1 – 3.6.3 for more information on multiple project activities). Where a grouped project includes multiple project activities, the project description shall designate which project activities may occur in each geographic area. Since all project instances are in Nepal thus fulfils this requirement.

Above has been verified by audit team during the site visit and submitted project documents. The same criteria to be checked during future inclusion of instances also.

EKI Energy Services Limited is responsible for project development. Legal ownership is held by of Asian Hydropower and Samling Power limited for installation of project. EKI Energy Services Limited is authorized PP by all owners of hydro power plant. As per the same, all the carbon credits related rights are with EKIESL; and EKIESL will be the concerned authority for all modalities and procedure with respect to VCUs of the project activity. The ownership of VCUs will be with EKIESL

The commissioning certificate for project activity are the supporting documents to demonstrate the project ownership.

Further as per agreement between Urja Developers Private Limited and EKI the VCUs rights are transferred. Samling Power Company Ltd. & Asian Hydropower Ltd. has authorized Urja Developers Private Limited and Urja Developers Private Limited has transferred all rights of carbon credit to EKIESL. The commissioning certificate for project activity are the supporting documents to demonstrate the project ownership.

This demonstrates the right of use according to clause 3.7.111.1 (3) of VCS Standard (v4.5) – “a project ownership arising by virtue of a statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals (where the project proponent has not been divested of such project ownership)”.

Validation & Verification Team has assessed and validated that the project start date for this project activity is 17-Septemernr-2021 as per the submitted objective evidence.

Project activity crediting period is following 17-Septemeber-2021 (Start date) to 16-Septemeber02028 (End date) Grouped project activity is identified as “Project scale” whereas it is estimated to remove GHG emissions reduction of between 20,000 – 100,000 tCO₂e/year as per the Section no 3.10.1 of VCS standard version 4.5.

The Geographical location of Grouped project activity is given below in the Table:

Table 3.1.1 Summary of location for project instances under scope of validation and verification

Project Identity	Installation Capacity (MW)	Physical Location	Latitude	Longitude
MBHP	9.51	Mai-Jogmai Municipality and Illam Municipality, Illam District, State-1, Eastern Region, Nepal	26°55'35.75"N	87°56'52.82"E
LJKHP	6.20	Jogami River of Illam District, Mechi Zone, Eastern Development Region, Nepal	26°53'48.37"N	87°59'31.58"E



Figure 3.1.1 Map of project activity instance containing details of plants

Validation & Verification Team has evaluated and found it accurate that proposed project activity is a greenfield project activity whereas in the baseline (as per applied methodology) equivalent amount of electricity would have been generated through Intensive connected Indian grid which further contributes to rise of GHG emissions in the atmosphere. As the Nepal power grid is connected to Indian grid which is fossil fuel intensive whereas Nepal grid

is not so the EF is take zero however, baseline scenario is justified as per India grid consideration in accordance with the methodology.

Validation & Verification Team found the response appropriate that during the project development phase and commissioning process respective requirement of laws, statutes are fulfilled by project activity instances. Also, there are no additional requirements on implementation of specified technology apart from commissioning as per National guidelines.

Validation & Verification Team conforms that proposed grouped project activity neither has been registered/seeking registration under any other GHG programs nor it is has been rejected under any other GHG programs. Also, Validation & Verification Team conforms that initial project activity instances are neither registered/being part of project activity in any other GHG program nor it has been rejected by any other GHG programs.

Validation & Verification Team conforms that neither the proposed project activity is not participating in any other Emission trading program and other binding limits nor the project activity instances participating in any other Emission trading program and other binding limits hence, VCU's claimed under current monitoring are not used as carbon credit claims/GHG's emission reduction claims under any other mechanism.

Validation and Verification Team conforms that proposed grouped project activity neither has generate any other form of GHG related environmental credit for GHG emission reductions/removals claimed under the VCS Program nor initial project activity instances have generate any other form of GHG related environmental credit for GHG emission reductions/ removals claimed under the VCS Program. Also, PP has not applied for scope 3 emissions under the supply chain of prosed project activity grouped project activity and associate project activity instance hence, PP is not obligated to report scope 3 emissions as per the requirement of VCS standard version 4.5.

Validation & Verifications team has assessed and concluded that there are no leakage emissions associated with the proposed hydro project activity as per the requirement of applied AMS.I.D, Version 18. Also, no commercially sensitive information is reported in PD-MR by PP under the scope of joint validation and verification also no other additional information provided in PD-MR which might be useful apart from regular evidence as pe VCS standard version 4.5 or/and applied methodology.

Validation & Verification Team has assessed the contribution of grouped project activity toward Sustainable Development Goals during current monitoring period and expected during the complete project cycle (for now first crediting period) described in PD-MR for SDG Target 7.2, 8 and 13.

Table 4.1.2 Assessment of grouped project for contribution toward SDGs

SDG Target	SDG Indicator	Conclusion of VVB
7.2	7.2.1 Renewable energy share in the total final energy consumption	VVB has assessed and concluded that the grouped project activity has supplied 91678.73 MWh during current monitoring period and it is supposed to supply the 525115 MWh for first crediting period.
8	8.6.1 Proportion of youth (aged 15-24 years) not in education, employment or training	VVB has assessed and concluded that the grouped project activity has contributed toward training program scheduled twice (2 Nos.) during current monitoring period.
13	13.2.2 Total Tonnes of greenhouse gas emissions per year avoided	VVB has assessed and concluded that the grouped project activity has avoided release of 26,567 tCO ₂ e during current monitoring period and it is supposed to avoid 152,187 tCO ₂ e for first crediting period.

Validation & Verification Team conforms that PP has provided the project description in accordance with the PD-MR template latest requirement where information is accurate and well described upon the nature of project and implementation statues is found in accordance with the project description. Validation & Verification Team has assessed the relevant assumption associated with estimated GHG emission reduction or removals which might vary from estimate one as project is having hydro power technology for energy generation which might vary depending upon natural cause resulting into flow pattern of water body as per seasonal variation.

3.2 Participation under Other GHG Programs

Validation & Verification Team conforms that project is not registered under any other approved GHG program hence, no gap validation or/and eligibility assessment in such context required for this project activity.

3.3 Safeguards

3.3.1 No Net Harm

Validation & Verification Team assessed that proposed grouped project activity promotes clean energy technology for which project emissions are considered zero as per applied methodology AMS.I.D, Version 18 also no environmental or social risk were identified which could cause any harm to environment. Same is validated during Local Stakeholder interviews during on-site assessment and concluded based upon the feedback received that the grouped project activity does not cause any net harm.

3.3.2 Local Stakeholder Consultation

As per the VCS requirements, it is necessary to invite the relevant stakeholders, prior of the validation process. Moreover, the stakeholder meeting took place prior to the start date of the project activity which fulfil the requirement of VCS standard 4.3. The Validation & Verification Team checked the relevance of the dates during the on-site assessment. The details of meeting are as given below:

Venue: Ward no. 10 of Ilam Municipality of Ilam District, Nepal.

Date of meeting: 17-March-2019 (Samling Power Company Ltd.) and 06-February-2019 (By Asian Hydropower Ltd.)

Feedback received by Samling Power Company Ltd. during LSC:

- During the implementation of the project, employment opportunities should be provided to skilled and unskilled workers at the local level.

The local people would be preferred for both skilled and unskilled work and same has been confirmed by VVB during interaction with project proponent and local stakeholders during site visit and MoM of the meeting.

- there will be any accident due to the project activity e.g. electric shock etc.

The chance of the accident in the project activity is negligible and PP would regularly check the safety measure as per the company policy to avoid such accident. Same has been confirmed by VVB during interaction with project proponent and local stakeholders during site visit and MoM of the meeting. Will there be free supply of power to the local people?

Since households are connected with local state electricity board and power from project activity will be exported to grid, thus technically direct power supply to the local people not possible but PP shall ensure that the availability of power to be used local people first and also increase the reliability of power supply. Free or any concession on the tariff rate of power would be local administration decision and PP do not have any control over the same as power of the project activity will be exported to grid. Same has been confirmed by VVB during interaction with project proponent and local stakeholders during site visit and MoM of the meeting

- Will it impact the climatic condition of the nearby area?. Climate condition of the nearby area will not impact due to project activity as clean technology (i.e. hydro power) used. PP has also taken all necessary approval as per local compliance requirement for the project. Same has been confirmed by VVB during interaction with project proponent and local stakeholders during site visit and MoM of the meeting

Feedback received by Asian Hydropower Ltd. during LSC:

- The project will have to build a road.

Project proponent will be built and repaired the road if anything required or damage due project activity during construction and entire operational period. Same has been confirmed by VVB during interaction with project proponent and local stakeholders during site visit and MoM of the meeting

- Should get proper compensation for the property.

Proper compensation as per local regulation should be given to property owner in case any property or land acquired for the project activity. Same has been confirmed by VVB during interaction with project proponent and local stakeholders during site visit and MoM of the meeting

- Locals should be given employment

The local people would be preferred for both skilled and unskilled work and same has been confirmed by VVB during interaction with project proponent and local stakeholders during site visit and MoM of the meeting

Validation & Verification Team has evaluated the response to the Q&A session conducted and confirms that PP has addressed the relevance of project activity to local stakeholders and seeking positive/negative feedback on development and implementation of project activity on their daily life. Validation & Verification Team has also interviewed some of the Local Stakeholder and validated the PD-MR information given in LSC section was found authentic.

Apart from above, project proponent was listed for stakeholder comments as per project standard 4.5 and no comments received during stakeholder comments period.

Further, ongoing communication with local stakeholders, local peoples are informed about the grievance register placed at site. If any concerns are received during the operation of the project activity, the same will be addressed if relevant to the project activity. For the current monitoring period no grievances have been received and same has been checked by audit team with grievances register during site visit.

3.3.3 Environmental Impact

According to the regulation, project activities conducted conduct Initial Environmental Examination (IEE) submitted to Ministry of Energy, Water Resources and Irrigation through Department of Electricity Development (DoED) in the month of March 2019 with Letter no. 075/076. The analysis of those impacts is done and mitigation measures are identified as per IEE report and same is included in EMP.

1. Physical Impact – The project activities like excavation and slope cutting in project components like powerhouse, penstock pipe, and surge pipe will lead to change in land topography. Impact on the topography will also occur by the clearance of land.

Mitigation Measures – Change in land use due to construction of structural components of the project like headworks, access road and switchyard permanent camp for operation workers is unavoidable and remains as residual impact till the project exists. Only the changes due to project support facilities can be mitigated.

Land area required for the other support project features such as labour camp, storage yard will be taken on lease.

2. Biological Impacts – Impact on surrounding forest resources for its products like firewood, timber and NTFPs.

Mitigation Measures – In order to minimize the need of forest products, the project workers will be provided kerosene to meet their fuel requirements for cooking purpose. However, emphasize will be given to provide LPG to the construction workers for cooking purpose.

The project management and contractor will impose complete prohibition on the purchase, sale and storage of forest products. Anyone found with local forest products in the camps will be penalized. This is a management task and no extra cost is envisaged

Validation & Verification Team has assessed IEE report and confirmed that environmental impacts of this project activity are negative. Even all mitigation plan against identified environment risk is being followed at the site and it is confirmed by audit team during site visit and also checked all necessary approval and clearance for both project instances and confirmed projects are in operation following all local regulations.

3.3.4 Public Comments

Validation & Verification Team conforms that no public comments are received to be received by VVB/PP during period of 30 days for Global Stakeholder Consultation on VERRA official website.

3.3.5 AFOLU-Specific Safeguards

Validation & Verification Team conforms that proposed project activity does not fall under AFOLU category hence, VVB is not required to assess the compliance of PD-MR for this section against the requirement of VCS standard version 4.5.

3.4 Application of Methodology

3.4.1 Title and Reference

As per the requirement of VCS standard version 4.5 PP has applied the below methodology to this project activity which is found valid under the scope of joint validation and verification:

Title- AMS-I.D.: Grid connected renewable electricity generation (Version 18.0)

URL Reference-

<<https://cdm.unfccc.int/methodologies/DB/W3TINZ7KKWCK7L8WTXFQQOFQQH4SBK> >

Tools-

- Tool to calculate the emission factor for an electricity system, Version 7.0
- Tool for the demonstration and assessment of additionality, Version 13.1
- Tool for Investment Analysis, Version 12

3.4.2 Applicability

Validation & Verification Team has assessed response against the criteria of AMS I.D, Version 18.0 for the proposed grouped project activity, below is the evaluation result and conclusion for respective criteria.

Table 3.4.2.1 Assessment of applicability conditions as per applied methodology and tools

S.N.	Applicability Condition of AMS I.D, Version 18.0	Conclusion of Validation & Verification Team
1.	This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)	Validation & Verification Team has assessed that proposed project activity falls under category (a) as there was not project activity existed operating before that start of this project activity as per the feasibility report. Hence, it concludes that applicability condition is met by proposed grouped project activity.

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| <p>2. Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology:</p> <p>(a)The project activity is implemented in an existing reservoir with no change in the volume of reservoir</p> <p>(b)The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4 W/m²</p> <p>(c)The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 W/m²</p> | <p>Validation & Verification Team has assessed that proposed project activity falls under a run of river. Hence, it concludes that applicability condition is not applicable to proposed grouped project activity.</p> |
| <p>3 If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.</p> | <p>Validation & Verification Team has assessed that proposed project activity does not have any non-renewable component, it is categorized as a greenfield. Hence, it concludes that applicability condition is not applicable to proposed grouped project activity</p> |
| <p>4 Combined heat and power (co-generation) systems are not eligible under this category.</p> | <p>Validation & Verification Team has assessed that proposed project activity does not consist of Combined heat and power (co-generation) systems. Hence, it concludes that applicability condition is not applicable to proposed grouped project activity.</p> |
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| <p>5. In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct¹ from the existing units.</p> | <p>Validation & Verification Team has assessed and concluded that there is no capacity addition of renewable energy generation units at an existing renewable power generation facility hence, criteria is not applicable.</p> |
| <p>6. In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW</p> | <p>Validation & Verification Team has assessed that proposed project activity is not having case of retrofit, rehabilitation or replacement. Hence, it concludes that applicability condition is not applicable to proposed grouped project activity.</p> |
| <p>7. In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as “AMS-I.C.: Thermal energy production with or without electricity” shall be explored</p> | <p>Validation & Verification Team has assessed that proposed project activity is not having case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions. Hence, it concludes that applicability condition is not applicable to proposed grouped project activity.</p> |
| <p>8. In case biomass is sourced from dedicated plantations, the applicability criteria in the tool “Project emissions from cultivation of biomass” shall apply</p> | <p>Validation & Verification Team has assessed that proposed project activity is not having case biomass sourcing from dedicated plantation. Hence, it concludes that applicability condition is</p> |
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		not applicable to proposed grouped project activity.
S.N.	Applicability Condition of Tool to calculate the emission factor for an electricity system	Conclusion of Validation & Verification Team
1.	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g., demand-side energy efficiency projects).	Validation & Verification Team has assessed and concluded that grouped project activity is supplying electricity to national grid hence, the latest version of this tool is applied.
2.	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off grid power plants. In the latter case, the conditions specified in "Appendix 2: Procedures related to off grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in	Validation & Verification Team has assessed and concluded that grouped project activity is supplying electricity to national grid hence, the latest version of this tool is applied for emission factor is calculated in accordance with the same.

	generation and not to other aspects such as transmission capacity.	
3.	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Validation & Verification Team has assessed and concluded that grouped project activity is not located in Annex I country neither partially nor totally hence, criteria in not applicable.
4.	Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	Validation & Verification Team has assessed and concluded that grouped project activity does not involve any biofuel hence the criteria is not applicable.

S.N.	Applicability Condition of Tool for the demonstration and assessment of additionality	Conclusion of Validation & Verification Team
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1.	The use of the methodological tool “Demonstration of additionality of small-scale project activities” is not mandatory for project participants when proposing new methodologies. Project participants and coordinating/managing entities may propose alternative methods to demonstrate additionality for consideration by the Executive Board.	Validation & Verification Team has assessed PP has chosen to apply the tool.
2.	Project participants and coordinating/managing entities may also apply “TOOL19: Demonstration of additionality of microscale project activities” as applicable.	Validation & Verification Team has assessed and concluded that PP has not applied the tool as it is not valid for this grouped project activity.

S.N.	Applicability Condition of Tool for Investment analysis	Conclusion of Validation & Verification Team
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| <p>1. This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the guidelines “Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.</p> | <p>Validation & Verification Team has assessed and concluded that PP applied the tool in accordance with the guidance under “Tool for the demonstration and assessment of additionality”.</p> |
| <p>2. In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.</p> | <p>Validation & Verification Team has assessed and concluded that PP applied the tool in accordance with the applied methodology.</p> |
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3.4.3 Project Boundary

Validation & Verification Team has assessed and confirms that below is the accurate information upon geographical location of respective project activity instances which are found consistent with respect to reporting in PD-MR and submitted commissioning certificates same are verified during the physical site visit.

Table 3.4.3.1 Summary of location for project instances under scope of validation and verification

Project Identity	Installation Capacity (MW)	Physical Location
MBHP	9.51	Mai-Jogmai Rural Municipality and Illam Municipality, Illam District, State-1, Eastern Region, Nepal
LJKHP	6.2	Jogami River of Ilam District, Mechi Zone, Eastern Development Region, Nepal

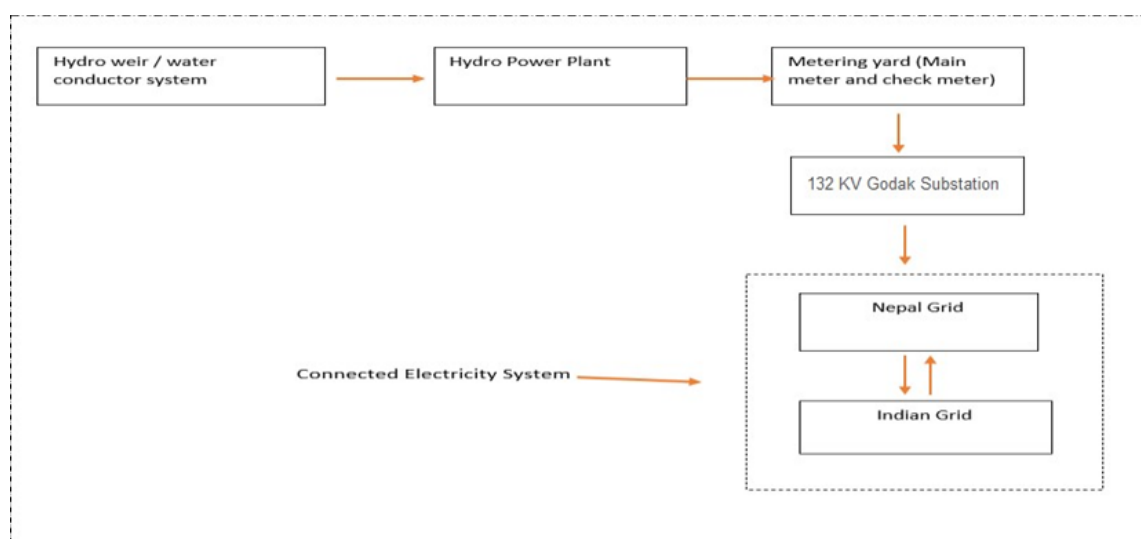


Figure 3.4.3.1 Project Boundary of grouped project

Validation & Verification Team has assessed and conforms that PP has identified the GHG source, sink and reservoir in accordance with the applied methodology AMS-I.D, Version 18 in accordance with VCS standard version 4.5 for both baseline and project also, no additional GHG source, sink and reservoir are identified during on site assessment by Validation & Verification Team.

Table 3.4.3.2 GHG source, sink or reservoir identified for grouped project activity

Source	Gas	Included?	Justification/Explanation
Baseline	CO2	Yes	Main emission source
	CH4	NO	Minor emission source

Source	Gas	Included?	Justification/Explanation
Project	Grid connected electricity generation	N2O NO	Minor emission source
		CO2 NO	No CO2 emissions are emitted from the project
	Greenfield Hydro Power Project activity	CH4 NO	Project activity does not emit CH4
		N2O NO	Project activity does not emit N2O

3.4.4 Baseline Scenario

Validation & Verification Team has assessed and conforms that as per applied methodology AMS-I.D, Version 18 in accordance with the VCS standard version 4.5, baseline scenario for this grouped project activity is fossil fuel based electricity generation via grid-connected power plant in Nepal (connected to Indian Grid) which is accurately and with reasonable assumptions described in respective section of PD-MR in accordance with commissioning certificates, PPAs. Same information is further confirmed by Validation & Verification Team with inserted power ministry of India link in the updated PDMR (version 05) which confirms Nepal is importing power from India as well as same has been confirmed further through the official website of National Electricity Authorities of Nepal⁹.

3.4.5 Additionality

Validation and Verification Team has assessed that PP has demonstrated upon the two components of additionally requirement in accordance with the 'Guidelines on the demonstration of Additionality of small-scale Project Activity' (version 9.0):

Legal Requirement Test

As the project is not mandate by enforced laws, statutes, regulations, court orders, environmental-mitigation agreements, permitting conditions of other legally-binding

⁹ Nepal Electricity Authority Annual Report 2021-22_ https://www.nea.org.np/annual_report

mandates hence, voluntary commitments/agreements for this particular sector for PP do not constitute the any legal requirement to be tested.

The investment decision date of both instances as per below

Sr. No.	Investor Name	Investment Decision Date	Commissioning Date
1.	Samling Power Company Ltd.	18-March-2019	17-September-2021
2.	Asian Hydropower Ltd.	18-March-2019	01-November-2021

Above dates has been confirmed by audit team with decision date evidence of the projects i.e investment approval note.

Additionality Assessment

Validation & Verification Team has assessed the project technology qualification under positive list and concluded that the following grid-connected renewable electricity generation technologies of installed is automatically additional 'TOOL21- Methodological tool: Demonstration of additionality of small-scale project activities Version 10.0 Section 5 para 11(a) and Tool 32-Methodological tool -positive list of technologies- Version 4.0..

a) Tool 21

(i) Solar technologies (photovoltaic and solar thermal electricity generation); (ii) Off-shore wind technologies; (iii) Marine technologies (wave, tidal); (iv) Building-integrated wind turbines or household rooftop wind turbines of a size up to 100 kW;

b) Tool 32

a) Solar thermal electricity generation including concentrating solar power ; (b) Off-shore wind technologies; (c) Marine wave technologies; (d) Marine tidal technologies; (e) Building-integrated wind turbines or household rooftop wind turbines of a size up to 100 kW; (f) Biomass internal gasification combined cycle As per the above criteria the grouped project activity is not identified as automatically additional.

For establishment of additionally below barriers are to be assessed as per Guidelines on the Demonstration of Additionality of Small-scale Project Activities (Ver. 13.1 EB 105 Annex 4):

- **Investment barrier**
- **Technological barrier**
- **Barrier analysis Other barriers**

PP has proposed investment barrier to demonstrate that without financial support from carbon revenue from VCS program it would have been finally less attractive where the Tool for Investment Analysis, Version 12 is used following benchmarking.

Benchmark selection & Financial Indicator:

Validation & Verification Team has assessed that PP has chosen financial indication option 1 which is in accordance with the Methodological Tool for Investment Analysis version 12.

Determine appropriate analysis method:

Validation & Verification Team has assessed that PP has chosen Option- III for benchmark analysis as an appropriate analysis method to demonstrate the investment barrier.

Benchmark Calculation

The cost of equity is determined by selecting the values provided in the Appendix, i.e., Default values for cost of equity (expected return on equity) is presented below:

Appendix of the CDM TOOL 27: Investment Analysis Version 12 specifies default value of expected return on equity in real terms for Energy Industries (Group 1) in Nepal = 12.62%.

The benchmark has been computed in the following manner:

$$\text{Nominal Benchmark} = \{(1 + \text{Real Benchmark}) * (1 + \text{Inflation rate})\} - 1$$

Where,

Real Benchmark = Default Value, i.e., 12.62% (as per Appendix of Annex 2, EB 116)

Inflation rate = Projected Inflation Forecast is 4.60%. Inflation forecast is taken from IMF data source as in Nepal central bank is not publishing the same. This is as per para 16 of TOOL 27: Investment Analysis Version 12, hence found accepted.

Benchmark estimation:

As per latest updated version 12 of Tool 27: Investment analysis tool, default value of expected return on equity in real terms for Energy Industries (Group 1) in Nepal is = 12.62%.

The date of decision making is 18-March-2019 of the project activity. Decision making date has been considered from the date of PP's Capex Approval Note represent approval from PP's board of director for approval to start expenditure for the project activity. Same has been acceptable to VVB team as in-line with the para 10 of Tool 27: Investment Tool version 12.0.

It is noticed that during decision making version 8.0 of Tool 27 was available and applicable to project activity. However, PP has referred the version 12 which was available at the time of RFR submission. Benchmark is calculated as 20.73% if version 8.0 is applied, however

benchmark is 17.80% is calculated with version 12. Lower Benchmark is conservative, hence applying version 12 is conservative approaches hence can be accepted.

Default Value Benchmark:

For Samling Power Company Ltd.

Benchmark Calculations	Value	Sources Link	Document Date
Default Value for Nepal as per UNFCCC guidelines	12.62%	https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf	02-November-2022
Inflation forecast (WPI Mean) as per IMF for 10yrs	4.60%	https://www.imf.org/en/Countries/NPL	01-March-2019
Benchmark (with 10yrs Forecast)	17.80%	Calculated	
Benchmark	17.80%	Calculated	

For Asian Hydropower Ltd.

Benchmark Calculations	Value	Sources Link	Document Date
Default Value for Nepal as per UNFCCC guidelines	12.62%	https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf	02-November-2022
Inflation forecast (WPI Mean) as per IMF for 10yrs	4.60%	https://www.imf.org/en/Countries/NPL	01-March-2019
Benchmark (with 10yrs Forecast)	17.80%	Calculated	
Benchmark	17.80%	Calculated	

The Post tax Equity IRR is evaluated for the entire lifetime of the project activity, i.e., 30 years. It is calculated based on the cash outflows from and cash inflows into the project activity. The IRR and Benchmark analysis are calculated in excel spreadsheet and same is submitted to the VVB during validation of project activity. Based on result of IRR excel spreadsheets, equity IRR is less than Benchmark. This substantiates that the investment is not financially attractive (Equity IRR for the project activity is less than the Benchmark). Thus, it can be easily concluded that project activity is additional & is not business as usual

scenario. Input values used in all investment analysis shall be valid and applicable at the time of the investment decision taken by the project proponent which can be clearly verified; thus, it complies with guidance 10 of EB 116, Annex 2.

DPR has been submitted to verification team. The DPR was available during decision making and financial profitability of the project was decided based on this DPR. Assessment team checked the DPR of the project activity and found that consideration of the project cost in VCS joint PD and MR is correct and it is in line with Appendix of Methodological tool 27: “Investment Analysis” Version 12.0, as per there investment decision date as well as in compliance to VCS Standard v4.5. Hence, the project cost consideration is justified. Assessment team checked the actual project cost and still the project do not breach the benchmark. The sensitivity analysis below confirms the same. Since the actual cost is considered, there is no way the cost can go up and thus the same is assessed to be correct. The profitability of the project, which forms the basis for IRR calculation is based on installed capacity, PLF, tariff rate, O&M cost, depreciation, and taxation.

The financial indicators for Samling Power company ltd are mentioned below:

For Samling Power Company Ltd.

Details of the project		VVB Assessment
State where the project is situated	Ilam District Province 1	The location of project is checked with the kml file provided by the client which is found accurate.
Total Capacity (MW)	9.51	The capacity of the situated in Ilam District Province 1, Nepal is 9.51 MW and was confirmed from the DPR page no. 15 prepared by Hydro tunnelling and research pvt. ltd on January 2019 which is found to be appropriate. Further, the capacity of the project was cross verified from the commissioning certificate and found to be accurate with the assumption made for investment analysis.
Date of Commissioning	17-Sep-21	Commissioning date for turbines involved in project was confirmed from the commissioning certificate and found to be appropriate.
Life of the plant (Yrs.)	30	Lifetime of project is sourced from DPR prepared by Hydro tunnelling and research pvt. ltd in January 2019. The lifespan was cross verified from a published research article in Elsevier Journal “ Evolution and future prospects of hydropower sector in Nepal: A review - ScienceDirect ”, which is found to be appropriate by the verification team.
Generation of electricity		VVB Assessment
PLF (%)	54.51%	PLF value was confirmed from DPR prepared by Hydro tunnelling and research pvt. ltd in January 2019 which is found to be appropriate.

Annual generation (kWh)	4,54,10,973	Annual electricity generation in kWh is calculated multiplying total capacity, PLF and number of hours in a year (8760) and found to be appropriate.
Tariff rate at the decision making (INR/kWh)	2.45	Tariff rate has been checked from the DPR prepared by Hydro tunnelling and research pvt. ltd on January 2019 which is found to be appropriate. As per PPA, the actual tariff is INR/kWh 2.45 (INR/kWh) which is the same as assumption, hence, found to be acceptable.
Operation and maintenance cost and Insurance		VVB Assessment
O & M Expenses (INR Mn.)	11.87	O&M expense has been checked from DPR prepared by Hydro tunnelling and research pvt. ltd on January 2019. The value is also cross verified by the O&M agreement provided by PP which is found to be appropriate. Hence, assumption provided in the DPR was accepted by the validation team.
Escalation in the operational expenses (%)	3.00%	O&M escalation per annum has been checked from DPR prepared by Hydro tunnelling and research pvt. ltd on January 2019 which is found to be appropriate. Hence, assumption provided in the DPR was accepted by the validation team.
Insurance (INR Mn.)	3.56	As per DPR prepared by Hydro tunnelling and research pvt. ltd on January 2019
Financial parameters		VVB Assessment
TOTAL COST (INR Mn.)	1,187.90	Total cost has been checked from DPR prepared by Hydro tunnelling and research pvt. ltd on January 2019. The actual cost is higher than the assumption. Hence, assumption provided in the DPR was accepted by the validation team.
Loan Amount (INR Mn.)	890.92	As per DPR prepared by Hydro tunnelling and research pvt. ltd on January 2019
Equity Investment (INR Mn.)	296.97	Project is 25% equity as checked from the DPR prepared by Hydro tunnelling and research pvt. ltd on January 2019. Project capital cost has been cross verified from the balance sheet and found that the investment is 25% equity. Hence, the assumption was found acceptable.
Term loan		VVB Assessment
Loan Amount (INR Mn.)	890.92	As per DPR prepared by Hydro tunnelling and research pvt. ltd on January 2019. Loan amount has been cross verified from the loan agreement and found that the loan amount considered for IRR is correct. Hence it is accepted by the verification team.
Interest rate (%)	11.50%	As per DPR prepared by Hydro tunnelling and research pvt. ltd on January 2019

Loan Tenure (Qtr.)	40	As per DPR prepared by Hydro tunnelling and research pvt. ltd on January 2019
Repayment Period (Qtr.)	40	As per DPR prepared by Hydro tunnelling and research pvt. ltd on January 2019
Repayment instalments value (INR Mn.)	22.273	Calculated Value
Book Depreciation (SLM Method)		VVB Assessment
Land	-	
Gross Depreciable Value (INR Mn.)	1,187.90	Calculated Value
Salvage Value (%)	10.00%	PP has considered 30 years of project operational lifetime as period of assessment. Hence, 10% of salvage value considered by the PP is reasonable and accepted by the team.
Salvage value (INR Mn.)	118.79	Calculated Value
Net Depreciable Value (INR Mn.)	1,069.11	Calculated Value
Residual Value (INR Mn.)	118.79	Calculated Value
IT Depreciation		VVB Assessment
IT Depreciation (%)	30.00%	As per Income Tax act
Income Tax		VVB Assessment
Financial Year	FY 2019-20	
Income tax rate (%)	20.00%	As Per Income Tax Rule
Corporate Tax / MAT (%)	25.00%	As per Income Tax rule
GST (%)	13.00%	As Per Income Tax Rule
Surcharge (%)	20.00%	As Per Income Tax Rule
Education cess (%)	4.00%	As Per Income Tax Rule
Final Tax rates		VVB Assessment
Income tax rate (%)	24.96%	Calculated Value

MAT (%)	31.20%	Calculated Value
GST (%)	13.52%	Calculated Value

For Asian Hydropower Ltd.

Details of the project		VVB Assessment
State where the project is situated	Ilam District Province 1	The location of project is checked with the kml file provided by the client which is found accurate.
Total Capacity (MW)	6.20	The capacity of the situated in Ilam District Province 1, Nepal is 6.20 MW and was confirmed from the DPR page no. 6 prepared by Infra Partners Pvt. Ltd on December 2018 which is found to be appropriate. Further, the capacity of the project was cross verified from the commissioning certificate and found to be accurate with the assumption made for investment analysis.
Date of Commissioning	1-Nov-21	Commissioning date for turbines involved in project was confirmed from the commissioning certificate and found to be appropriate.
Life of the plant (Yrs.)	30	Lifetime of project is sourced from DPR prepared by Hydro tunnelling and research pvt. ltd in December 2018. The lifespan was cross verified from a published research article in Elsevier Journal " Evolution and future prospects of hydropower sector in Nepal: A review - ScienceDirect ", which is found to be appropriate by the verification team.
Generation of electricity		VVB Assessment
PLF (%)	54.51%	PLF value was confirmed from DPR prepared by Infra Partners Pvt. Ltd on December 2018 which is found to be appropriate.
Annual generation (kWh)	2,96,05,471	Annual electricity generation in kWh is calculated multiplying total capacity, PLF and number of hours in a year (8760) and found to be appropriate.
Tariff rate at the decision making (INR/kWh)	2.45	Tariff rate has been checked from the DPR prepared by Infra Partners Pvt. Ltd on December 2018 which is found to be appropriate. As per PPA, the actual tariff is INR/kWh 2.45 (INR/kWh) which is the same as assumption, hence, found to be acceptable.
Operation and maintenance cost and Insurance		VVB Assessment
O & M Expenses (INR Mn.)	7.39	O&M expense has been checked from DPR prepared by Infra Partners Pvt. Ltd on December 2018. The value is also cross verified by the O&M agreement provided by

		PP which is found to be appropriate. Hence, assumption provided in the DPR was accepted by the validation team.
Escalation in the operational expenses (%)	3.00%	O&M escalation per annum has been checked from DPR prepared by Infra Partners Pvt. Ltd on December 2018 which is found to be appropriate. Hence, assumption provided in the DPR is not breaching the limit which was accepted by the validation team.
Insurance (INR Mn.)	2.22	As per DPR prepared by Infra Partners Pvt. Ltd on December 2018.
Financial parameters		VVB Assessment
TOTAL COST (INR Mn.)	739.14	Total cost has been checked from DPR prepared by Infra Partners Pvt. Ltd on December 2018. The actual cost is higher than the assumption. Hence, assumption provided in the DPR was accepted by the validation team.
Loan Amount (INR Mn.)	554.36	As per DPR prepared by Infra Partners Pvt. Ltd on December 2018.
Equity Investment (INR Mn.)	184.79	Project is 25% equity as checked from the DPR prepared by Infra Partners Pvt. Ltd on December 2018. Project capital cost has been cross verified from the balance sheet and found that the investment is 25% equity. Hence, the assumption was found acceptable.
Term loan		VVB Assessment
Loan Amount (INR Mn.)	554.36	As per DPR prepared by Infra Partners Pvt. Ltd on December 2018. Loan amount has been cross verified from the loan agreement and found that the loan amount is correct. Hence it is accepted by the verification team.
Interest rate (%)	11.50%	As per DPR prepared by Infra Partners Pvt. Ltd on December 2018
Loan Tenure (Qtr.)	40	As per DPR prepared by Infra Partners Pvt. Ltd on December 2018
Repayment Period (Qtr.)	40	As per DPR prepared by Infra Partners Pvt. Ltd on December 2018
Repayment instalments value (INR Mn.)	13.859	Calculated Value
Book Depreciation (SLM Method)		VVB Assessment
Land	-	
Gross Depreciable Value (INR Mn.)	739.14	Calculated Value

Salvage Value (%)	10.00%	PP has already considered 30 years of project operational lifetime as period of assessment. Hence, 10% of salvage value considered by the PP is reasonable and accepted by the team.
Salvage value (INR Mn.)	73.91	Calculated Value
Net Depreciable Value (INR Mn.)	665.23	Calculated Value
Residual Value (INR Mn.)	73.91	Calculated Value
IT Depreciation		WVB Assessment
IT Depreciation (%)	30.00%	As per Income Tax act
Income Tax		WVB Assessment
Financial Year	FY 2019-20	
Income tax rate (%)	20.00%	As Per Income Tax Rule
Corporate Tax / MAT (%)	25.00%	As per Income Tax rule
GST (%)	13.00%	As Per Income Tax Rule
Surcharge (%)	20.00%	As Per Income Tax Rule
Education cess (%)	4.00%	As Per Income Tax Rule
Final Tax rates		WVB Assessment
Income tax rate (%)	24.96%	Calculated Value
MAT (%)	31.20%	Calculated Value
GST (%)	13.52%	Calculated Value

The project activity is a renewable source of electricity generation and supplies the electricity to the Nepal Electricity grid. The key parameters which determine the Equity IRR of the project activity are project cost, PLF Tariff and O&M cost.

IRR of both projects are calculated as below:

Sr. No.		Equity IRR	Benchmark (Equity IRR)
1	Samling Power Company Ltd.	3.21%	17.80%
2	Asain Hydropower Ltd.	3.74%	17.80%

Parameters and assumptions used and cross checking:

Assessment of project cost:

In the Joint PD& MR, the estimation of project cost is based on DPR available at time of Approval Note (investment decision making), The details of Input Parameters considered in IRR are as below:

Estimated Project Cost (In Million INR) – Asian Hydro Power limited	DPR Date
739.14	01-December-2018

As the project is already implement and actual project cost is available to cross check. Actual project cost of Asian Hydro Power limited is 657.4 Million INR which is lower than the cost considered at the time of decision making. IRR at actual project cost is 5.4% which is still below the benchmark. Hence project is additional at actual project cost. Actual project is verified from the Purchase orders.

Estimated Project Cost (In Million INR) – Samling Power company limited	DPR Date
1187.90	01-Jan-2019

As the project is already implement and actual project cost is available to cross check. Actual project cost of Samling Power company limited is 1227.77Million INR which higher than the cost considered at the time of decision making. Hence project is additional at actual project cost. Actual project is verified from the purchase orders.

Assessment of Plant Load Factor (PLF):

The PLF is considered as mentioned in Detailed Project Report (DPR) which was available at the time of investment decision. The same was prepared by the third-party company Infra partners pvt ltd for Asian Hydropower pvt ltd and by Hydro Tunnelling & research pvt ltd for Samling Power company limited. Hence the value considered by the project owner for demonstrating additionality of the project is deemed acceptable to the assessment team and also in line with paragraph 3 (b) of “Guidelines for the reporting and Validation of Plant Load Factors” (Annex 11 of EB 48).

Estimated PLF – Asian Hydro Power limited	DPR Date
54.51%	01-December-2018

As the project is already commissioned actual PLF is available to cross check the PLF taken at the time of decision making. Actual PLF is evaluated from actual generation in the Monitoring period. Actual PLF is 62% and IRR is 5.89% at this PLF. IRR at actual PLF is well below that benchmark, Project is not breaching the benchmark at actual PLF.

Estimated PLF – Samling Power company limited	DPR Date
54.51%	01-Jan-2019

As the project is already commissioned actual PLF is available to cross check the PLF taken at the time of decision making. Actual PLF is evaluated from actual generation in the Monitoring period. Actual PLF is 64% and IRR is 5.9% at this PLF. IRR at actual PLF is well below that benchmark, Project is not breaching the benchmark at actual PLF.

Assessment of Tariff:

Estimated tariff is considered as mentioned in DPR which was available at the time of investment decision:

Estimated tariff – Asian Hydro Power limited	DPR Date
2.454 INR/kWh	01-December-2018

As the project is already implement and actual Tariff is available to cross check. Actual Tariff is cross checked from PPA and Tariff i.e. power purchase price is 2.88 INR/kWh. IRR will breach the benchmark if Tariff exceeds 4.79 INR/kWh. As the PPA is signed for project life time hence there is not chance to increase Tariff further. Hence project is not breaching the benchmark as actual Tariff.

Estimated tariff – Samling Power company limited	DPR Date
2.454 Rs/kWh	01-Jan-2019

As the project is already implement and actual Tariff is available to cross check. Actual Tariff is cross checked from PPA and Tariff i.e. power purchase price is 2.88 INR/kWh. IRR will breach the benchmark if Tariff exceeds 4.79 INR/kWh. As the PPA is signed for project life time hence there is not chance to increase Tariff further. Hence project is not breaching the benchmark as actual Tariff.

Assessment of O& M cost:

The DPR has been used in the financial calculation as same was available during decision making and hence applicable. According to Appendix of Methodological tool “Investment Analysis” at time of investment decision making, the cost should be based on the input parameters available at the time of decision making and the PP has submitted offer letter supporting this consideration. Therefore, considering the above assessment, verification team concluded that the O&M cost considered from respective offer letter in the computation of financial indicator is in conformity with guidance Appendix of Methodological tool “Investment Analysis” Version 12.

Estimated O&M (In Million INR) – Asian Hydro Power limited	DPR Date
7.39	01-December-2018

As the project is already implement and actual O&M cost is available to cross check. Actual O&M is cross checked form O&M agreement between Asian Hydro Power limited with Urja Services Pvt Ltd and actual O&M is 6.768 Million INR. IRR at actual O&M cost is 3.94% which is still below the benchmark. Hence project is additional at actual O&M cost.

Estimated O&M (In Million INR) – Samling Power company limited	DPR Date
11.87	01-Jan-2019

As the project is already implement and actual O&M cost is available to cross check. Actual O&M is cross checked form O&M agreement between Samling Power company limited with Urja Services Pvt Ltd and actual O&M is 7.199 Million INR. IRR at actual O&M cost is 4.19% which is still below the benchmark. Hence project is additional at actual O&M cost.

Sensitivity Analysis

Validation and Verification Team has assessed the demonstration upon project cost, Tariff, O&M and PLF. Equity IRR Project is not breaching with actual value of any of parameters as demonstrated above. Sensitivity analysis is carried out at +/-10% of each input parameter. Results of sensitivity analysis are presented as below;

For Samling Power Company Ltd.

Sensitivity Analysis	Equity IRR			
Variation %	-10%	Normal	10%	Breaching Value
PLF	1.63%	3.21%	4.77%	100.20%
O&M	3.47%	3.21%	2.96%	-728.50%
Project Cost	4.67%	3.21%	2.01%	-53.75%

Tariff Rate	1.63%	3.21%	4.77%	100.20%
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For Asain Hydropower Ltd.

Sensitivity Analysis	Equity IRR			
Variation %	-10%	Normal	10%	Breaching Value
PLF	2.14%	3.74%	5.31%	95.38%
O&M	3.97%	3.74%	3.50%	-727.00%
Project Cost	5.22%	3.74%	2.51%	-52.33%
Tariff Rate	2.14%	3.74%	5.31%	95.38%

The results of sensitivity analysis show that even with a variation of +10% & 10% in project cost, O&M cost, PLF and Power cost, Equity IRR is significantly lower than the benchmark. And it is evident from the results given above; the project remains additional even under the most favourable conditions.

Validation & Verification Team conforms that determined IRR is less than Benchmark values in all the possible scenario, hence, conclude that the proposed project activity is less financially attractive as per sensitivity analysis.

Barrier analysis

Validation & Verification Team has assessed that barrier analysis is not proposed by PP.

Common practice analysis

Validation & Verification Team has assessed that common practice analysis is not applicable to this small-scale project activity.

Hence, Validation & Verification has found demonstration of PP in line with the requirements of additionality tool applied titled as 'Demonstration of additionality of small-scale project activities' version 13.1 upon additionality aspects.

3.4.6 Quantification of GHG Emission Reductions and Removals

Baseline Emissions are calculated by PP as below:

As per AMS I.D. version 18-

$$BE_y = EG_{PJ,y} \times EF_{grid,y}$$

Where

BE_y : Baseline emissions in year y (t CO₂)

$EG_{PJ,y}$: Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in project year (MWh)

$EF_{grid,y}$ Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO₂/MWh)

Project Emissions are calculated by PP as below:

As per AMS I.D. version 18

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y}$$

Where,

PE_y : Project emissions in year y (t CO₂e/yr)

$PE_{FF,y}$: Project emissions from fossil fuel consumption in year y (t CO₂/yr)

$PE_{GP,y}$: Project emissions from the operation of dry, flash steam or binary geothermal power plants in year y (t CO₂e/yr)

$PE_{HP,y}$: Project emissions from water reservoirs of hydro power plants in year y (t CO₂e/yr)

As per the applied methodology of the project activity, project emission to be considered in case the power density is less than 10 W/m². However as demonstrated in section 5.2 of PDMR, in both project instances, the power density is greater than 10 W/m², project emission in both instances is nil. The input values used in the density calculation has been verified by audit team with evidences and found correct.

As a group project and instances can be included in future, thus power density criteria must be checked as per applied methodology is required during each instance inclusion time.

Leakage Emissions calculated by PP:

Validation & Verification Team has assessed that as per the requirement of AMS I. D version 18 PP is not required to calculate leakage emission for this grouped project activity as no biomass residues is produced.

Validation & Verification Team has assessed and conforms that PP has followed for approach for quantification of GHG emission reductions and removal in line with applied methodology AMS I D Version 18

$$ER_y = BE_y - PE_y - LE_y$$

Where,

ER_y = Emission reductions in year y (t CO₂/y)

BE_y = Baseline Emissions in year y (t CO₂/y)

PE_y = Project emissions in year y (t CO₂/y)

LE_y = Leakage emissions in year y (t CO₂/y)

BE_y = 21,741 (t CO₂/y) and for the first crediting period total baseline emission reductions are estimated to be 152,187 tCO₂ reported by PP (refer to Estimate ER sheet)

PE_y = 0 as per AMS I.D. version 18

LE_y = 0 as per AMS I.D. version 18

Table 3.4.6.1 Assessment of parameters at validation as per applied methodology and tools

Validation & Verification Team has assessed and concluded that and emission factor of Nepal electricity system is not available as it is not published by Nepal Electricity Authority (NEA). Hence PP has taken the emission factor of Nepal is considered as zero from IFI Default Grid Factors 2021 v3.2 published by UNFCCC. However, Nepal is importing electricity from India. Approximately 32% electricity is imported from Indian grid and emission factor of Indian grid is calculated as 0.9105 by using CEA database as per TOOL07. Hence the emission factor of Nepal is calculated as weighted average emission factor of Nepal and India. Asian Development Bank has published the emission factor of Nepal is 348 tCO₂/ GWh . Weighted average of Nepal and India grid is taken as 0.2898 tCO₂/ MWh (i.e. 289 tCO₂/ GWh) which is lower than Asian Development Bank and should be considered as a conservative

Parameter at validation	Description of Parameter	Unit	Value Applied	Conclusion of VVB
$EF_{grid, OM, y}$ (Indian Grid)	Operating Margin CO ₂ emission factor in year y	tCO ₂ e/ MWh	0.9524	Determined as per Baseline CO ₂ Emission Database published by CEA (Version 18) in accordance with CDM TOOL07
$EF_{grid, BM, y}$ (Indian Grid)	Build Margin CO ₂ emission factor in year y	tCO ₂ e/ MWh	0.8687	Determined as per Baseline CO ₂ Emission Database published by CEA (Version 18) in accordance with CDM TOOL07
$EF_{grid, CM, y}$ (Indian Grid)	Combined Margin CO ₂ emission factor in year y	tCO ₂ e/ MWh	0.9105	Determined as per Baseline CO ₂ Emission Database published by CEA (Version 18)

					in accordance with CDM TOOL07
$EF_{grid, CM, y}$ (Nepal Grid)	Combined Margin CO ₂ emission factor in year y	tCO ₂ e/ MWh	0		Applied as per Harmonized IFI Default Grid Factors 2021 v3.2 in accordance with CDM TOOL07
Parameter to be monitored	Description of Parameter	Unit	Value Applied	Conclusion of VVB	
$EG_{PJ, y}$	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y	MWh/y	75,016 (at validation), 91,678.73 (at first verification)	Estimated at validation stage using PLF and Installation Capacity which is to be monitored on monthly basis at verification cycle in accordance with AMS I.D. Version 18	

Assessment team conclude that the above-mentioned baseline scenario is in line with the applied methodology “AMS I.D.: Grid Connected Renewable Electricity Generation, version 18.0”. Same is also reflected in final Joint PDMR. Thus accepted.

The combined margin emissions factor is calculated as follows:

$$EF_{grid, CM, y} = EF_{grid, OM, y} \times w_{OM} + EF_{grid, BM, y} \times w_{BM}$$

Where:

$EF_{grid, BM, y}$ = Build margin CO₂ emission factor in year y (t CO₂/MWh)

$EF_{grid, OM, y}$ = Operating margin CO₂ emission factor in year y (t CO₂/MWh)

w_{OM} = Weighting of operating margin emissions factor (per cent)

w_{BM} = Weighting of build margin emissions factor (per cent)

Therefore,

$$EF_{grid, CM, y \text{ India}} = 0.9524 \times 0.5 + 0.8687 \times 0.5 \\ = \mathbf{0.9105 \text{ tCO}_2/\text{MWh}}$$

- Combined Margin Emission Factor for Nepal:

$$EF_{grid, CM, y \text{ Nepal}} = 0.0 \text{ tCO}_2/\text{MWh}^{10}$$

¹⁰ <https://unfccc.int/documents/461676>

Nepal Electricity Generation ¹¹	6052000	MWh
Power Purchase from India ¹²	2826000	MWh

- Connected system weighted average combined Emission Factor:

$$\frac{\text{Nepal Generation} * EF_{\text{grid,CM,y, Nepal}} + \text{Power purchase from India} * EF_{\text{grid,CM, y, India}}}{\text{Nepal Electricity Generation} + \text{Power Purchase from India}}$$

$$= 6052000 \times 0 + 2826000 \times 0.9105 / (6052000 + 2826000)$$

Connected system weighted average combined Emission Factor = 0.2898 tCO₂/MWh.

3.4.7 Methodology Deviations

Validation & Verification Team has assessed and concluded that there is not methodological deviation identified against the requirement of VCS standard version 4.5.

3.4.8 Monitoring Plan

Validation & Verification Team has assessed the monitoring plan developed and implemented by PP with respect to responsible personnel monitoring and measurement, quality assurance, quality control procedures as described below:

Organogram of team responsible for operation and measurement:

¹¹ https://www.nea.org.np/annual_report

¹² https://www.nea.org.np/annual_report

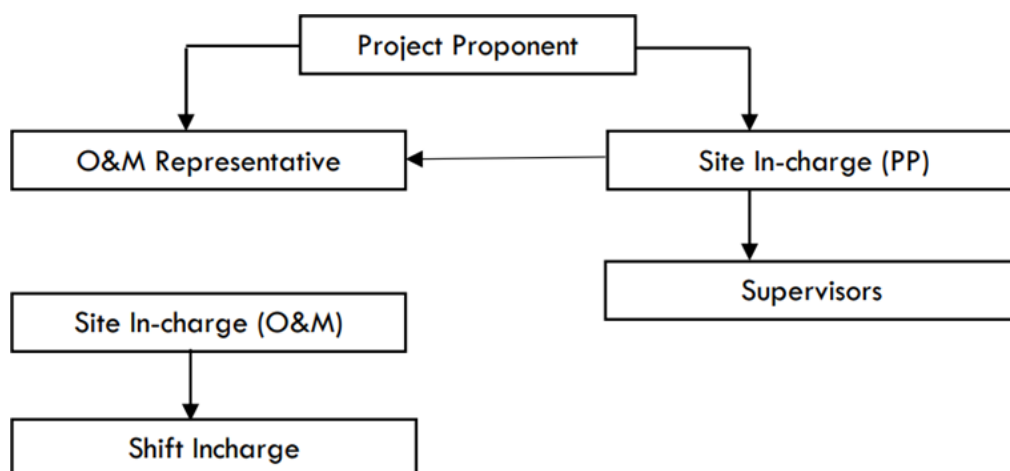


Figure 3.4.8.1 Organogram of personnel responsible of operation & maintenance

Table 3.4.8.1 Assessment of Monitoring Plan as per applied methodology

Function	Reporting Tasks	Recording	Conclusion of VVB
Data Measurement	Electricity Import/Export to grid	Meter reading notation on daily basis, invoice to be manged	Joint Meter reading as well as respective invoices are crosschecked and found valid
Data collection and archiving	Manual and electronic recording of import and export to grid	Logbook is maintained	Logbook is crosschecked with Joint meter reading report and found valid
QA/QC Procedures	Calibration of meters at decided intervals	Calibration certificates obtained	Calibration Certificate, validity, result is checked against applied methodological requirement of AMS I.D. version 18
Personnel training	Operation and Maintenance Training on	Training schedule and certificates if available	Training schedule is checked and interview of respective personnel taken during onsite assessment.

regular interval				
Parameter to be monitored	Description of Parameter	Unit	Value Applied	Conclusion of VVB
EG _{PJ, y}	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y	MWh/y	75,016 (at validation), 91,678.73 (at first verification)	Estimated at validation stage using PLF and Installation Capacity which is to be monitored on monthly basis at each verification cycle in accordance with AMS I.D. Version 18

3.5 Non-Permanence Risk Analysis

Validation & Verification Team has assessed and concluded that PP is not required to perform Non-performance risk analysis against the requirement of VCS standard version 4.5.

4 VERIFICATION FINDINGS

4.1 Accuracy of GHG Emission Reduction and Removal Calculations

Baseline Emissions are calculated by PP as below:

As per AMS I.D. version 18-

$$BE_y = EG_{PJ,y} \times EF_{grid,y}$$

Where

BE_y: Baseline emissions in year y (t CO₂)

EG_{PJ,y}: Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in project year (MWh)

EF_{grid,y} Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO₂/MWh)

Project Emissions are calculated by PP as below:

As per AMS I.D. version 18

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y}$$

Where,

PE_y : Project emissions in year y (t CO₂e/yr)

$PE_{FF,y}$: Project emissions from fossil fuel consumption in year y (t CO₂/yr)

$PE_{GP,y}$: Project emissions from the operation of dry, flash steam or binary geothermal power plants in year y (t CO₂e/yr)

$PE_{HP,y}$: Project emissions from water reservoirs of hydro power plants in year y (t CO₂e/yr)

1. Samling Power Company Ltd. -

The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation $\left(PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}} \right)$, is greater than 10 W/m²

$$\frac{9510000 \text{ W}}{18250 \text{ m}^2}$$

$$= 521.09 \text{ W/m}^2$$

As the calculated value from the given equation is 521.09 W/m² which is greater than 10 W/m², so as per applied methodology for the project activity, there would not be any project emission.

2. Asian Hydropower Ltd. –

The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation $\left(PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}} \right)$, is greater than 10 W/m²

$$\frac{6200000 \text{ W}}{18250 \text{ m}^2}$$

$$= 339.72 \text{ W/m}^2$$

As the calculated value from the given equation is 339.72 W/m² which is greater than 10 W/m², so as per applied methodology for the project activity, there would not be any project emission.

Leakage Emissions calculated by PP:

Validation & Verification Team has assessed that as per the requirement of AMS I. D version 18 PP is not required to calculate leakage emission for this grouped project activity as no biomass residues is produced.

Validation & Verification Team has assessed and conforms that PP has followed for approach for quantification of GHG emission reductions and removal in line with applied methodology AMS I D Version 18.

$$ER_y = BE_y - PE_y - LE_y$$

Where,

ER_y : Emission reductions in year y (t CO₂/y)

BE_y : Baseline Emissions in year y (t CO₂/y)

PE_y : Project emissions in year y (t CO₂/y)

LE_y : Leakage emissions in year y (t CO₂/y)

BE_y : 26,567 tCO₂ for first verification period (17-September-2021 to 31-December-2022) reported by PP (refer to Actual ER sheet) whereas estimated emission reduction is 28,055 tCO_{2e} for the same duration

$PE_y = 0$ as per AMS I.D. version 18

$LE_y = 0$ as per AMS I.D. version 18

Table 4.1.1 Assessment of data and parameter at verification as per applied methodology and tool

Parameter at validation	Description of Parameter	Unit	Value Applied	Conclusion of VVB
$EF_{grid, OM, y}$ (Indian Grid)	Operating Margin CO ₂ emission factor in year y	tCO _{2e} /MWh	0.9524	Determined as per Baseline CO ₂ Emission Database published by CEA (Version 18) in accordance with CDM TOOL07
$EF_{grid, BM, y}$ (Indian Grid)	Build Margin CO ₂ emission factor in year y	tCO _{2e} /MWh	0.8687	Determined as per Baseline CO ₂ Emission Database published by CEA (Version 18) in accordance with CDM TOOL07
$EF_{grid, CM, y}$ (Indian Grid)	Combined Margin CO ₂ emission factor in year y	tCO _{2e} /MWh	0.9105	Determined as per Baseline CO ₂ Emission Database published by CEA (Version 18)

in accordance with CDM TOOL07

Equation:

$$EF_{grid,CM,y} = EF_{grid,OM,y} * W_{OM} + EF_{grid,BM,y} * W_{BM}$$

$EF_{grid,CM,y}$ Combined Margin $tCO_2e/$ 0
(Nepal Grid) CO_2 emission MWh
factor in year y

Applied as per Harmonized IFI Default Grid Factors 2021 v3.2 in accordance with CDM TOOL07

Where Connected system weighted average combined Emission Factor = 0.2898 tCO_2e/MWh (Refer to PD-MR)

Parameter to be monitored	Description of Parameter	Unit	Value Applied	Conclusion of VVB
$EG_{PJ,y}$	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y	MWh/y	75,016 (at validation), 91,678.73 (at first verification)	Estimated at validation stage using PLF and Installation Capacity which is monitored on monthly basis at first verification cycle in accordance with AMS I.D. Version 18

4.2 Quality of Evidence to Determine GHG Emission Reductions and Removals

Validation & Verification Team as verified the ER sheet and confirms that estimated and actual emission reductions or/and removals are calculated using the unitary method. Validation & Verification Team has assessed the commissioning certificates, equipment/instrument specifications, Power Purchase Agreements, Contractual evidence, logbook for daily electricity measurement, invoices for electricity import/export under scope

of first verification during desk review as well as on-site visit if possible. Also, Validation & Verification Team has interviewed respective stakeholders to conform accuracy of data/parameter(s) value(s), reasonableness and adequacy of the fact/assumption(s) and validity of objective evidence (records/public references) against the requirement of VCS standard version 4.5.

5 VALIDATION AND VERIFICATION OPINION

TÜV SÜD South Asia Pvt. Ltd. has successfully completed performed Joint validation and verification of the “GROUPED HYDRO POWER PROJECT BY EKI ENERGY SERVICES LIMITED” by Urja Developers Private Limited and EKI Energy Services Limited.

The management of the project participant/owner is responsible for the preparation of the GHG emissions data and the reported/estimated GHG emissions reductions on the basis set out within the project’s Monitoring Plan in the Joint VCS PD-MR and the approved CDM methodology AMS I.D. version 18.

TÜV SÜD’s Validation & Verification approach is based on the requirements of VCS rules in accordance with the latest VCS standard 4.5 which is risk-based, drawing upon an understanding of the risks identified and addressed for estimated/actual GHG emissions data quantified or/and to be quantified. TÜV SÜD South Asia Pvt. Ltd. conforms that:

- Project Implementation of following grouped project activity is complying with the VCS rules and requirements.
- Demonstration of baseline scenario of following grouped project activity is in accordance with the applied VCS rules and methodological requirements
- Monitoring and measurement plan is developed and implemented in accordance with the applied VCS rules and methodological requirements
- Control of objective evidence is in accordance applied VCS rules and methodological requirements
- Quantification of GHG emission reductions or/and removals are calculated and reported without material misstatement by PP and assessed by VVB at reasonable level of assurance (materiality threshold <5%).

The grouped project activity is expected to reduce 152,187 tCO₂ during the length of first crediting period from 17-September-2021 to 16-September-2028 (first and last date included).

Verification period: 17-September-2021 to 31-December-2022 (first and last date included)

Table 5.1.1 Estimated GHG emission reductions or removals (tCO₂e) for first crediting period

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
17-September-2021 to 16-September -2022	21,741
17-September-2022 to 16-September -2023	21,741
17-September-2023 to 16-September -2024	21,741
17-September-2024 to 16-September -2025	21,741
17-September-2025 to 16-September -2026	21,741
17-September-2026 to 16-September -2027	21,741
17-September-2027 to 16-September -2028	21,741
Total estimated ERs	152,187
Total number of crediting years	7
Average annual ERs	21,741

Table 5.1.2 Net GHG emission reductions or removals (tCO₂e) for first verification period

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
17-September-2021 to 31-December-2021	2,239	0	0	2,239
01-January-2022 to 31-	24,328	0	0	24,328

December-2022				
Total	26,567	0	0	26,567

Table 5.1.3 Comparison between Estimated and Achieved emissions reductions/removals for first crediting period

Year	Ex-ante emissions reductions /removals	Achieved emissions reductions /removals	Percent difference	Justification for the difference
17-September-2021 to 31-December-2021	8,165	2,239	-72.58	Water availability is subject to seasonal variation (dependency upon rainfall intensity), Weather conditions (cyclones/typhons) other human/anthropogenic activities (blockage due to trash) over which PP holds no control
01-January-2022 to 31-December-2022	28,105	24,328	-13.44	Water availability is subject to seasonal variation (dependency upon rainfall intensity), Weather conditions (cyclones/typhons) other human/anthropogenic activities (blockage due to trash) over which PP holds no control
Total	28,055	26,567	-5.30	Water availability is subject to seasonal variation (dependency upon rainfall intensity), Weather conditions (cyclones/typhons) other human/anthropogenic activities (blockage due to trash) over which PP holds no control

APPENDIX:

<1→LIST OF FINDINGS>

Table 5. Remaining FAR from validation and/or previous verifications

Not Applicable.

Table 6. CL from this validation and verification

CL ID	00	Section no.	1.1, 1.12, 2.2. 2.3, 3.5, 4.1	Date:	05-July-2023
Description of CL					
PP is requested to provide below listed supporting documents:					
1. Power Purchase Agreement for Mai Beni Hydropower Project (MBHP) and Jogmai Khola Hydropower Project (LJKHP) Section 1.1 2. KML file Section 1.12 3. LSC Report (Valid) Section 2.2 4. Feasibility Reports (Authorized) Section 2.3 5. IRR Sheet (Information and Calculation Approach for combined IRR sheet) Section 3.5 6. Proof of Technical Specifications Section 4.1					
Project participant response					Date: 10-July-2023
1. Power Purchase Agreement for Mai Beni Hydropower Project (MBHP) and Jogmai Khola Hydropower Project (LJKHP) has now been provided. 2. KML file has now been provided. 3. LSC Report has now been provided. 4. Feasibility Reports has now been provided. 5. IRR Sheet has now been provided. 6. Technical Specifications has now been provided.					
Documentation provided by project participant					
All supporting documents					
VVB assessment					Date: 11-July-2023

PP is requested to provide below listed supporting documents:	
1. Translated version of Power Purchase Agreement for Mai Beni Hydropower Project (MBHP) and Jogmai Khola Hydropower Project (LJKHP) Section 1.1 2. Training records Section 1.17.2 3. Translated version of public notice, attendance sheet Section 2.2 4. Authorized Feasibility Reports Section 2.3 5. Supplier agreement for equipment Section 3.5	
Project participant response	Date: 13-July-2023
1. Power Purchase Agreement for Mai Beni Hydropower Project (MBHP) and Jogmai Khola Hydropower Project (LJKHP) has already been provided. 2. Training records has already been provided. 3. MoM copy of English Translation has now been provided. 4. Authorized Feasibility Reports has now been provided. 5. Supplier agreement for equipment has now been provided.	
VVB assessment	Date: 13-JULY-2023
PP has incorporated the accurate information in the respective section hence, VVB accepts the response.	

Table 3. CL from this validation and verification

CL ID	01	Section no.	1.1	Date: 05-July-2023
Description of CL				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 1.1 of PD-MR, Project Instance details are given stating “EKI Energy Services Limited” customer. As it was noticed Asian Hydropower and Samling Power limited had authorized Urja Developers for entering into ERPA with EKI it is not clear which entity is responsible for operation hence PP needs to clarify role for “Urja Developers” and its own in the context of handling project activity operation.				
Project participant response				Date: 10-July-2023
Asian Hydropower and Samling Power limited has the legal owner of this project activity and same has been corrected in section 1.1 of the revised Joint PDMR. Urja Developers is the project proponent and EKI Energy Services Ltd. is also the project proponent of this project activity and owning the VCUs as account registry is on EKI name, you can check the same from communication agreement. EKI also play a role of consultant for this project activity.				
Documentation provided by project participant				
Revised Joint PDMR				
VVB assessment				Date: 11-July-2023

PP has provided the details of legal ownership, which is consistent with objective evidence hence, VVB accepts the response.

Table 4. CL from this validation and verification

CL ID	02	Section no.	1.4	Date: 05-July-2023
Description of CL				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In paragraph 1-4 of section 1.4 of PD-MR, PP has referred to outdated VCS standard version 4, 4.1 at certain places in order to demonstrated eligibility of project activity instances hence, PP needs to justify why it has it not referred the latest version of VCS Standard for the same?				
Project participant response				Date: 10-July-2023
Section 1.4 has now been updated				
Documentation provided by project participant				
Revised Joint PDMR				
VVB assessment				Date: 11-July-2023
PP has corrected the reference of VCS standard as per latest hence, VVB accepts the response.				

Table 5. CL from this validation and verification

CL ID	03	Section no.	1.4	Date: 05-July-2023
Description of CL				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In paragraph 4 of section 1.4 of PD-MR, PP has stated that “If any project activity instance is bundled project, then that project activity instance capacity is within capacity limit of 15 MW” however, the demonstration refers to outdated version of the VCS standard hence, PP need to justify how does it fulfill capacity limitation criteria requirement in accordance with the latest version of VCS standard?				
Project participant response				Date: 10-July-2023
Section 1.4 has now been updated				
Documentation provided by project participant				
Revised Joint PDMR				

VVB assessment	Date: 11-July-2023
In paragraph 07 of PD-MR PP has stated “As per para 3.6.9 of VCS Standard V4.4, each project activity instances are not exceeding the capacity limit of 15 MW.”, “Also, each project activity instances do not exceeding the 1% of capacity limit of 15 MW.” However, PP needs to provide clarity upon the total capacity limit as per applied methodology in accordance with the VCS standard where VCS standard version 4.4 clearly states “Where a capacity limit applies to a project activity included in the project, no project activity instance shall exceed such limit. Further, no single cluster of project activity instances shall exceed the capacity limit”	
Project participant response	Date: 13-July-2023
The capacity limit is 15 MW for small scale project activity. This project activity is group of number of projects where capacities are varying from each other. However, to meeting applicability criteria of small scale each project activity should not have capacity more than 15 MW.	

Table 6. CL from this validation and verification

CL ID	04	Section no.	1.5, 1.6	Date: 05-July-2023
Description of CL				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01 In section 1.5 and 1.6 of PD-MR, EKI Energy Services Limited is listed as PP and also, the other party hence PP needs to clarify why so as the section 1.6 requires information upon “any other entities involved in the development of the project” which is not the case here?				
Project participant response				Date: 10-July-2023
EKI Energy Services Ltd. is the project proponent of this project activity and owning the VCU as account registry is on EKI name, you can check the same from communication agreement. EKI also play a role of consultant for this project activity.				
Documentation provided by project participant				
NA				
VVB assessment				Date: 11-July-2023
PP has provided the appropriate justification hence, VVB accepts the response.				

Table 7. CL from this validation and verification

CL ID	05	Section no.	1.11	Date: 05-July-2023
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Description of CL	
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01	
In section 1.11 of PD-MR, PP has stated that “A power purchase agreement has also been signed between Samling Power Company and Nepal Electricity Authority” however, PPA for both project activity instances are not provided hence, PP needs to provide the same.	
Project participant response	Date: 10-July-2023
Power Purchase Agreement for Mai Beni Hydropower Project (MBHP) and Jogmai Khola Hydropower Project (LJKHP) has now been provided.	
Documentation provided by project participant	
PPA	
VVB assessment	Date: 11-July-2023
PP has provided the PPAs however, same are not provided in translated version for key information.	
Project participant response	Date: 13-July-2023
PPAs for Mai Beni Hydropower Project (MBHP) and Jogmai Khola Hydropower Project (LJKHP) has already been provided.	
VVB assessment	Date: 13-JULY-2023
PP has incorporated the accurate information in the respective section hence, VVB accepts the response.	

Table 8. CL from this validation and verification

CL ID	06	Section no.	1.12	Date: 05-July-2023
Description of CL				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 1.12 of PD-MR, PP has stated that “This is a grouped project activity and the Nepal is the project activity. The KML file shared with the DOE”, “A KML file have been submitted in order to indicate the project location and geographic boundaries” however, KML file is not provided hence, PP needs to provide the same.				
Project participant response				Date: 10-July-2023
KML file has now been provided				
Documentation provided by project participant				

KML file	
VVB assessment	Date: 11-July-2023
PP has provided the KML file hence, VVB accepts the response.	

Table 9. CL from this validation and verification

CL ID	07	Section no.	1.17.2	Date: 05-July-2023
Description of CL				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 1.17.2 of PD-MR, PP has only demonstrated 2 SDGs (Goals) which is not in accordance with the requirement of VCS Standard Version 4.4 as at least 3 SDGs to be demonstrated, PP needs to provide the clarity upon the same.				
Project participant response				Date: 10-July-2023
3 SDGs has now been provided in section 1.17.2 of the revised joint PDMR.				
Documentation provided by project participant				
Revised Joint PDMR				
VVB assessment				Date: 11-July-2023
PP has provided incorporated at least 3 SDGs and respective information however, training record does not contain sufficient information and found invalid as objective evidence for SDG indicator 8.6.1.				
Project participant response				Date: 13-July-2023
Training records has already been provided.				
VVB assessment				Date: 13-JULY-2023
PP has incorporated the accurate information in the respective section hence, VVB accepts the response.				

Table 10. CL from this validation and verification

CL ID	08	Section no.	2.2	Date: 05-July-2023
Description of CL				

Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01	
In section 2.2 of PD-MR, PP had to provide the respective detail as per objective evidence same is submitted however, the signature of attendees was found invalid hence, PP needs to provide the valid objective evidence.	
Project participant response	Date: 10-July-2023
Attendance copy has now been provided	
Documentation provided by project participant	
Attendance copy	
VVB assessment	Date: 11-July-2023
PP has provided the invitation and attendance sheet for LSC however, same is not provided in translated version.	
Project participant response	Date: 13-July-2023
MoM copy of English Translation has now been provided.	
VVB assessment	Date: 13-July-2023
PP has provided relevant objective evidence hence, PP accepts the response.	

Table 11. CL from this validation and verification

CL ID	09	Section no.	2.3	Date: 05-July-2023
Description of CL				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 2.3 of PD-MR, PP has not provided the authorized feasibility reports hence, PP needs to provide the valid objective evidence.				
Project participant response				Date: 10-July-2023
Feasibility report has now been provided				
Documentation provided by project participant				
Feasibility reports				
VVB assessment				Date: 11-July-2023
PP has provided the feasibility reports however, same are not authorized.				

Project participant response	Date: 13-July-2023
Authorized Feasibility Reports has now been provided.	
VVB assessment	Date: 13-JULY-2023
PP has provided the information hence, VVB accepts the response.	

Table 12. CL from this validation and verification

CL ID	10	Section no.	3.5	Date: 05-July-2023
Description of CL				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 3.5 of PD-MR, PP has provided objective evidence information however feasibility report contains analysis for individual project activity instance however, IRR sheet has one single analysis result hence, PP needs to provide complete information in objective evidence with calculation approach.				
Project participant response				Date: 10-July-2023
Separate IRR sheet for both of the project activity has now been provided.				
Documentation provided by project participant				
IRR sheet				
VVB assessment				Date: 11-July-2023
PP has provided the IRR sheets for both project activity instance hence, VVB accepts the response.				

Table 13. CL from this validation and verification

CL ID	11	Section no.	3.5	Date: 05-July-2023
Description of CL				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 3.5 of PD-MR, PP has not provided the objective evidence for technical specifications of equipment/instruments hence, PP needs to provide the same.				
Project participant response				Date: 10-July-2023
Technical specification sheet has now been provided				
Documentation provided by project participant				

Technical specifications	
VVB assessment	Date: 11-JULY-2023
PP has provided the technical specifications however agreement with the supplier is not provided.	
Project participant response	Date: 13-July-2023
Supplier agreement for equipment has now been provided.	
VVB assessment	Date: 13-JULY-2023
PP has provided the information hence, VVB accepts the response.	

Table 14. CL from this validation and verification

CL ID	12	Section no.	6.1	Date: 05-July-2023
Description of CL				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 6.1 of PD-MR, PP has not referred to “Data Source: Central Electricity Authority (CEA) database Version 17.0, October 2021” for determining the $EF_{grid,BM,y}$ value however, the source is outdated hence, PP needs to provide the same.				
Project participant response				Date: 10-July-2023
Version of Central Electricity Authority (CEA) database has been updated in section 5.1 of the revised Joint PDMR.				
Documentation provided by project participant				
Revised Joint PDMR				
VVB assessment				Date: 11-JULY-2023
PP has stated the latest source of information hence, VVB accepts the response.				

Table 15. CL from this validation and verification

CL ID	13	Section no.	6.1	Date: 05-July-2023
Description of CL				

Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01	
In section 6.1 of PD-MR, PP has provided the outdated reference for determining the parameters EF_{grid, OM,y} (Indian Grid) and EF_{grid, BM,y} (Indian Grid) value hence, PP needs to provide clarity upon the same.	
Project participant response	Date: 10-July-2023
Version of Central Electricity Authority (CEA) database has been updated in section 6.1 of the revised Joint PDMR.	
Documentation provided by project participant	
Revised Joint PDMR	
VVB assessment	Date: 11-JULY-2023
PP has stated the latest source of information hence, VVB accepts the response.	

Table 16. CL from this validation and verification

CL ID	14	Section no.	7.1	Date: 05-July-2023
Description of CL				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 7.1 of PD-MR, PP has not provided the monitored value for Parameter EG_{PJ, y} value for current monitoring period also net electricity generation zero for July 2021, August 2021, October 2022, November 2022 and Dec 2022 for 9.52 MW plant also, for October 2021, November 2021 and December 2021 it is negative? Why is the net electricity generation zero for September 2021, November 2022 and Dec 2022 for 6.2 MW plant in ER sheet hence, PP needs to provide justification upon the same.				
Project participant response				Date: 10-July-2023
Monitored value has now been provided in section 7.1 of the revised PDMR. Generation is zero for July 2021, August 2021, October 2022, November 2022 and Dec 2022 for 9.51 MW and September 2021, November 2022 and Dec 2022 for 6.2 MW plant as per JMR.				
Documentation provided by project participant				
Revised Joint PDMR				
VVB assessment				Date: 11-JULY-2023
PP has provided the monitored value for the parameter however, response to the finding is not containing sufficient justification with respect to zero generation in specified months.				

Project participant response	Date: 13-July-2023
Justification with respect to zero generation has been mentioned in JMR and pending months JMRs for October 2022, November 2022 and Dec 2022 has now been provided.	
VVB assessment	Date: 13-JULY-2023
PP has incorporated the accurate information in the respective section hence, VVB accepts the response.	

Table 17. CL from this validation and verification

CL ID	15	Section no.	3.5	Date: 11-July-2023
Description of CL				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 3.5 of PD-MR, PP has stated that "At the time of decision made of project activity, Version 05 of methodological tool "Investment Analysis" (EB 62, Annex 5) was the latest available tool to Project owner. However, the request for registration for Version 5 could be submitted till 23/07/2015. Hence, Project owner has used Methodological Tool for Investment Analysis version 11 (EB 112, Annex 02)." However, evidence of such decision-making timeline is lacking also, PP has to demonstrate why the latest tool requirements cannot be met or applied hence, PP need to provide clarity upon the same.				
Project participant response				Date: 13-JULY-2023
Latest tool of investment analysis has now been provided in section 3.5 of the revised joint PDMR.				
Documentation provided by project participant				
Revised Joint PDMR				
VVB assessment				Date: 11-JULY-2023
PP has corrected the information and incorporated in the respective section hence, VVB accepts the response.				

Table 18. CAR from this validation and verification

CAR ID	01	Section no.	1.1	Date: 05-July-2023
Description of CAR				

Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01	
In section 1.1 of PD-MR, Project Instance details are given stating Date of Commissioning for 6.2 MW plant as “01-October-2021” which is not consistent with the commissioning certificate hence, PP needs incorporate the accurate information in the respective section.	
Project participant response	Date: 10-July-2023
Date of Commissioning has been corrected in section 1.1 of the revised Joint PDMR.	
Documentation provided by project participant	
Revised Joint PDMR	
VVB assessment	Date: 11-JULY-2023
PP has not corrected the commercial operation date as per the commissioning certificate.	
Project participant response	Date: 13-JULY-2023
Commissioning date has now been corrected in section 1.1 of the revised Joint PDMR.	
VVB assessment	Date: 13-JULY-2023
PP has corrected the commercial operation date as per the commissioning certificate, hence PP accepts the response	

Table 19. CAR from this validation and verification

CAR ID	02	Section no.	1.1	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 1.1 of PD-MR, PP has not provided the details in project history table in accordance with the requirement of Joint Project Description and Monitoring Report, v4.2 template hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 10-July-2023
Details in project history table has now been provided in section 1.1 of the revised Joint PDMR.				
Documentation provided by project participant				
Revised Joint PDMR				
VVB assessment				Date: 11-JULY-2023
PP has provided the details in project history table hence VVB accepts the response.				

Table 20. CAR from this validation and verification

CAR ID	03	Section no.	1.3	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 1.3 of PD-MR, PP has not provided the sufficient information with respect to qualification under VCS program scope against the requirement of Joint Project Description and Monitoring Report, v4.2 template hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 10-July-2023
Section 1.3 has now been updated				
Documentation provided by project participant				
Revised Joint PDMR				
VVB assessment				Date: 11-JULY-2023
PP has provided the sufficient information upon eligibility of project, hence VVB accepts the response.				

Table 21. CAR from this validation and verification

CAR ID	04	Section no.	1.8	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 1.8 of PD-MR, PP has stated that “PP is willing to include the project activity instances during registration having commissioning dates from 01- August-2021 onward and as per deadline of 2 years of validation completion from date of commissioning” however, earliest start date is not found consistent with respective objective evidence hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 10-July-2023
Date of Commissioning has been corrected in section 1.8 of the revised Joint PDMR.				
Documentation provided by project participant				
Revised Joint PDMR				
VVB assessment				Date: 11-JULY-2023

PP has not corrected the project start date in accordance with the commissioning certificate.	
Project participant response	Date: 13-JULY-2023
Project start date has now been corrected in section 1.8 of the revised Joint PDMR.	
VVB assessment	Date: 13-JULY-2023
PP has corrected the project start date in accordance with the commissioning certificate, hence PP accepts the response.	

Table 22. CAR from this validation and verification

CAR ID	05	Section no.	1.17.2	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 1.17.2 of PD-MR, PP has not provided the sufficient information upon SD contribution of project activity against the requirement of Joint Project Description and Monitoring Report, v4.2 template hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 10-July-2023
Sufficient information has now been provided in section 1.17.2 of the revised Joint PDMR.				
Documentation provided by project participant				
Revised Joint PDMR				
VVB assessment				Date: 11-JULY-2023
PP has provided the sufficient information of contribution to SDGs hence, PP accepts the response.				

Table 23. CAR from this validation and verification

CAR ID	06	Section no.	1.17.2	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 1.17.2 of PD-MR, PP has not provided the accurate details of SDG indicator in the table hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 10-July-2023
Accurate details of SDG indicator have now been provided in section 1.17.2 of the revised Joint PDMR.				

Documentation provided by project participant	
Revised Joint PDMR	
VVB assessment	Date: 11-JULY-2023
PP has not provided the accurate information on respective SDG indicator hence, PP accepts the response.	

Table 24. CAR from this validation and verification

CAR ID	07	Section no.	1.17.2	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 1.17.2 of PD-MR, PP has not provided the accurate information against the column “Current Project Contributions” against the requirement of Joint Project Description and Monitoring Report, v4.2 template hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 10-July-2023
Information of Current Project Contributions has now been provided in section 1.17.2 of the revised Joint PDMR.				
Documentation provided by project participant				
Revised Joint PDMR				
VVB assessment				Date: 11-JULY-2023
PP has provided the accurate information to Current Project Contributions” hence, VVB accepts the response.				

Table 25. CAR from this validation and verification

CAR ID	08	Section no.	2.2	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 2.2 of PD-MR, PP has not provided the sufficient information against requirement of Joint Project Description and Monitoring Report, v4.2 template hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 10-July-2023

Section 2.2 has now been updated	
Documentation provided by project participant	
Revised Joint PDMR	
VVB assessment	Date: 11-JULY-2023
PP has provided sufficient information upon Local Stakeholder Consultation hence, VVB accepts the response.	

Table 26. CAR from this validation and verification

CAR ID	09	Section no.	2.3	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 2.3 of PD-MR, PP has not provided the summary of conclusion of Environmental Impact assessment against the requirement of Joint Project Description and Monitoring Report, v4.2 template hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 10-July-2023
Section 2.3 has now been updated				
Documentation provided by project participant				
Revised Joint PDMR				
VVB assessment				Date: 11-JULY-2023
PP has provided sufficient information upon Environmental Impact aspect hence, VVB accepts the response.				

Table 27. CAR from this validation and verification

CAR ID	10	Section no.	3.2	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 3.2 of PD-MR, PP has provided irrelevant information on applicability conditions against the requirement of applied methodology hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 10-July-2023

Section 3.2 has now been updated	
Documentation provided by project participant	
Revised Joint PDMR	
VVB assessment	Date: 11-JULY-2023
PP has eliminated the irrelevant information with respect to applicability conditions hence, VVB accepts the response.	

Table 28. CAR from this validation and verification

CAR ID	11	Section no.	3.2	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 3.2 of PD-MR, PP has provided Project activity applicability conditions in Table 2 however, no reference provided for the same hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 10-July-2023
Section 3.2 has now been updated				
Documentation provided by project participant				
Revised Joint PDMR				
VVB assessment				Date: 11-JULY-2023
PP has corrected the information in the section hence, VVB accepts the response.				

Table 29. CAR from this validation and verification

CAR ID	12	Section no.	3.2	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 3.2 of PD-MR, PP has provided response against Point 1. of the ‘Applicability conditions of “Tool to calculate the emission factor for an electricity system”’ stating “This condition is applicable. OM, BM and CM are estimated using the tool under section B.6.1 for calculating baseline emissions.” However, no section as B.6.1 is identified in the document hence, PP needs to incorporate the accurate information in the respective section.				

Project participant response	Date: 10-July-2023
Section 3.2 has now been updated	
Documentation provided by project participant	
Revised Joint PDMR	
VVB assessment	Date: 11-JULY-2023
PP has corrected the information of reference hence, VVB accepts the response.	

Table 30. CAR from this validation and verification

CAR ID	13	Section no.	3.2	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 3.2 of PD-MR, PP has provided response against Point 3. of the ‘Applicability conditions of “Tool to calculate the emission factor for an electricity system”’ stating “The project activity is located in Nepal, a non-Annex I country” However, no reference for the identification is provided hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 10-July-2023
Section 3.2 has now been updated				
Documentation provided by project participant				
Revised Joint PDMR				
VVB assessment				Date: 11-JULY-2023
PP has incorporated the reference to the source of information hence, VVB accepts the response.				

Table 31. CAR from this validation and verification

CAR ID	14	Section no.	3.2	Date: 05-July-2023
Description of CAR				

Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01	
In section 3.2 of PD-MR, PP has provided response against Point 4. of the ‘Applicability conditions of “Tool to calculate the emission factor for an electricity system”’ stating “The project activity is a grid connected hydro power project and not a hydro power plant. Therefore, this criterion is not applicable” However, invalid response is provided hence, PP needs to incorporate the accurate information in the respective section.	
Project participant response	Date: 10-July-2023
Section 3.2 has now been updated	
Documentation provided by project participant	
Revised Joint PDMR	
VVB assessment	Date: 11-JULY-2023
PP has corrected the information on project technology hence, VVB accepts the response.	

Table 32. CAR from this validation and verification

CAR ID	15	Section no.	3.2	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 3.2 of PD-MR, PP has stated that “Also, the total installed capacity of each project activity instance less than 15 MW threshold limit of small-scale project activity instances and respective methodology will be applicable for that project activity instances” However, same is not demonstrate against the requirement of applicable criteria of methodology and latest version of VCS standard hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 10-July-2023
Section 3.2 has now been updated				
Documentation provided by project participant				
Revised Joint PDMR				
VVB assessment				Date: 11-JULY-2023
PP has not demonstrated how is the capacity limit applied against the specified criteria of applied methodology in accordance with the VCS standard version 4.4?				
Project participant response				Date: 13-JULY-2023

The capacity limit is 15 MW for small scale project activity. This project activity is group of number of projects where capacities are varying from each other. However, to meeting applicability criteria of small scale each project activity should not have capacity more than 15 MW.	
VVB assessment	Date: 13-JULY-2023
PP has demonstrated that capacity limit applied against the specified criteria of applied methodology hence, VVB accepts the response.	

Table 33. CAR from this validation and verification

CAR ID	16	Section no.	3.3	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 3.3 of PD-MR, PP stated that “The project boundary is therefore the physical boundary, which includes diversion weir, intake chamber, de-silting chamber, power channel, forebay, headrace tunnel, penstock, powerhouse, tailrace, the transmission system to the evacuation point and auxiliary power consumed by the plant” However, the information is not consistent with project boundary schematic provided hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 10-July-2023
Information has now been corrected in section 3.3 of the revised joint PDMR.				
Documentation provided by project participant				
Revised Joint PDMR				
VVB assessment				Date: 11-JULY-2023
PP has corrected the information upon project boundary hence, VVB accepts the response.				

Table 34. CAR from this validation and verification

CAR ID	17	Section no.	3.3	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 3.3 of PD-MR, PP has provided the details on Indian grid which is identified as connected electricity system however the information is not consistent with section 1.1 hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 10-July-2023

Project boundary diagram has now been updated in section 3.3 of the revised joint PDMR.	
Documentation provided by project participant	
Revised Joint PDMR	
VVB assessment	Date: 11-JULY-2023
PP has not provided the grid system information in section 1.1 in accordance with the section 3.3.	
Project participant response	Date: 13-JULY-2023
Grid system information has now been provided in section 1.1 of the revised joint PDMR.	
VVB assessment	Date: 13-JULY-2023
PP has incorporated the accurate information in the respective section hence, VVB accepts the response.	

Table 35. CAR from this validation and verification

CAR ID	18	Section no.	3.5	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 3.5 of PD-MR, PP has not provided sufficient information against the requirement applied tool hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 10-July-2023
Sufficient information has now been provided in section 3.5 of the revised joint PDMR.				
Documentation provided by project participant				
Revised Joint PDMR				
VVB assessment				Date: 11-JULY-2023
PP has incorporated information as per applied tools for additionality hence, VVB accepts the responses.				

Table 36. CAR from this validation and verification

CAR ID	19	Section no.	4.1	Date: 05-July-2023
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Description of CAR	
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01	
In section 4.1 of PD-MR, PP has not provided sufficient information upon technical specifications of equipment/instrument hence, PP needs to incorporate the accurate information in the respective section.	
Project participant response	Date: 10-July-2023
Technical specification details have now been provided in section 4.1 of the revised joint PDMR.	
Documentation provided by project participant	
Revised Joint PDMR	
VVB assessment	Date: 11-JULY-2023
PP has provided the sufficient information upon technical specifications of equipment/instrument hence, VVB accepts the response.	

Table 37. CAR from this validation and verification

CAR ID	20	Section no.	5.1	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 5.1 of PD-MR, PP has not provided description of parameters $EG_{PJ,y}$ and $EF_{grid,y}$ as per the requirement of applied methodology hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 10-July-2023
Description has now been provided in section 5.1 of the revised joint PDMR.				
Documentation provided by project participant				
Revised Joint PDMR				
VVB assessment				Date: 11-JULY-2023
PP has provided the accurate description of parameters $EG_{PJ,y}$ and $EF_{grid,y}$ hence, VVB accepts the response.				

Table 38. CAR from this validation and verification

CAR ID	21	Section no.	5.1	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In paragraph 1-2 section 5.1 of PD-MR, PP has not provided the source of information hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 10-July-2023
Source of information has now been provided in section 5.1 of the revised joint PDMR.				
Documentation provided by project participant				
Revised Joint PDMR				
VVB assessment				Date: 11-JULY-2023
PP has not provided the source of information under step 1 paragraph 1-2.				
Project participant response				Date: 13-JULY-2023
Information has now been provided in section 5.1 of the revised joint PDMR.				
VVB assessment				Date: 13-JULY-2023
PP has incorporated the accurate information in the respective section hence, VVB accepts the response.				

Table 39. CAR from this validation and verification

CAR ID	22	Section no.	5.1	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 5.1 of PD-MR, PP has stated that "Hence, 10 % of the project electricity system is 145.13 MW. And thus, the transmission capacity of the transmission line(s) that is connecting electricity systems (700 MW) is more than 10 per cent of the installed capacity (145.13 MW)." However, objective evidence for the information is not provided hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 10-July-2023
Weblink has now been provided for the same.				

Documentation provided by project participant	
Revised Joint PDMR	
VVB assessment	Date: 11-JULY-2023
PP has incorporated the reference however same is found invalid link.	
Project participant response	Date: 13-JULY-2023
Valid weblink has now been updated for the same.	

Table 40. CAR from this validation and verification

CAR ID	23	Section no.	5.1	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 5.1 of PD-MR, PP has stated that "Power Purchase from India" However, objective evidence for the information is not provided hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 10-July-2023
Revised estimated ER sheet has been provided.				
Documentation provided by project participant				
Revised ER sheet				
VVB assessment				Date: 11-JULY-2023
PP has not addressed the finding as the objective evidence does not contain the amount of electricity imported/exported to India grid. Also, there was not change in ER sheet was observed with respect to this particular finding.				
Project participant response				Date: 13-JULY-2023
Weblink has now been provided and also cross check the amount of electricity imported/exported to India grid from Estimated ER sheet.				
VVB assessment				Date: 13-JULY-2023
PP has incorporated the accurate information in the respective section hence, VVB accepts the response.				

Table 41. CAR from this validation and verification

CAR ID	24	Section no.	5.1	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 5.1 of PD-MR, PP has determined the values of parameter “Connected system weighted average combined Emission Factor” however the approach for calculation/determination is not provided against the requirement of applied methodology hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 10-July-2023
Weblink has now been provided for the same.				
Documentation provided by project participant				
Revised Joint PDMR				
VVB assessment				Date: 11-JULY-2023
PP has not addressed the finding just revised the value of the parameter.				
Project participant response				Date: 13-JULY-2023
Calculation has been incorporated in section 5.1 of the revised joint PDMR.				
VVB assessment				Date: 13-JULY-2023
PP has incorporated the accurate information in the respective section hence, VVB accepts the response.				

Table 42. CAR from this validation and verification

CAR ID	25	Section no.	5.4	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 5.4 of PD-MR, PP has not provided accurate value of total estimate GHG Estimated baseline emissions or removals (tCO ₂ e) and Estimated net GHG emission reductions or removals(tCO ₂ e) hence, PP needs to incorporate the accurate information in the respective section and ER sheet.				
Project participant response				Date: 10-July-2023
Value has now been corrected in section 5.4 of the revised joint PDMR.				
Documentation provided by project participant				

Revised Joint PDMR	
VVB assessment	Date: 11-JULY-2023
PP has provided the accurate value of total estimate GHG Estimated baseline emissions or removals (tCO ₂ e) and Estimated net GHG emission reductions or removals (tCO ₂ e) in accordance with the revised ER sheet hence, VVB accepts the response.	

Table 43. CAR from this validation and verification

CAR ID	26	Section no.	6.2	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 6.2 of PD-MR, PP has not provided response against "Monitoring equipment" for the parameter EG_{PJ,y} hence, PP needs to incorporate the accurate information in the respective section and ER sheet.				
Project participant response				Date: 10-July-2023
Description of Monitoring equipment has now been provided in section 6.2 of the revised joint PDMR.				
Documentation provided by project participant				
Revised Joint PDMR				
VVB assessment				Date: 11-JULY-2023
PP has provided the information on monitoring equipment for EG_{PJ,y} hence, VVB accepts the response.				

Table 44. CAR from this validation and verification

CAR ID	27	Section no.	7.5	Date: 05-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 7.5 of PD-MR, PP has not provided any response against requirement of Joint Project Description and Monitoring Report, v4.2 template hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 10-July-2023
Response has now been corrected in section 7.5 of the revised joint PDMR.				
Documentation provided by project participant				
Revised Joint PDMR				

VVB assessment	Date: 11-JULY-2023
PP has not provided the calculation approach for yearly (including first day as per actual start date of project instance chosen which is not in accordance with commissioning certificate) values for Baseline GHG emission reductions or removals (tCO ₂ e) and Net GHG emission reductions or removals (tCO ₂ e).	
Project participant response	Date: 13-JULY-2023
Calculation has been incorporated in section 7.5 of the revised joint PDMR.	
VVB assessment	Date: 13-JULY-2023
PP has incorporated the accurate information in the respective section hence, VVB accepts the response.	

Table 45. CAR from this validation and verification

CAR ID	28	Section no.	1.17.2	Date: 11-July-2023
Description of CAR				
Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01				
In section 1.17.2 of PD-MR, PP has not provided valid response to “Contributions Over Project Lifetime” as it is same as “current project contribution” against the requirement of Joint Project Description and Monitoring Report, v4.2 template hence, PP needs to incorporate the accurate information in the respective section.				
Project participant response				Date: 13-JULY-2023
“Contributions Over Project Lifetime” and “current project contribution” value is same because this is the first verification of this project activity.				
Documentation provided by project participant				
NA				
VVB assessment				Date: DD-MM-YYYY
PP has incorporated the accurate information in the respective section hence, VVB accepts the response.				

Table 46. CAR from this validation and verification

CAR ID	29	Section no.	1.17.2	Date: 11-July-2023
Description of CAR				

Monitoring Period: 17-September-2021 to 31-December-2022, Crediting Period: 01	
In section 1.17.2 of PD-MR, PP has not provided accurate information on SDG target against the requirement of Joint Project Description and Monitoring Report, v4.2 template hence, PP needs to incorporate the accurate information in the respective section.	
Project participant response	Date: 13-JULY-2023
Accurate information on SDG target has been provided in section 1.17.2 of the revised joint PDMR.	
Documentation provided by project participant	
Revised Joint PDMR	
VVB assessment	Date: DD-MM-YYYY
PP has incorporated the accurate information in the respective section hence, VVB accepts the response.	

Table 47. FAR for this Joint validation and verification

Not Available.

<2→ LIST OF DOCUMENTS REVIEWED>

No.	Author	Title	References to the document
1	EKI Energy Services Limited & Urja Developers Private Limited	VCS 4139_PD-MR_V03_TA-R2-Response	Refer to attachment
2	EKI Energy Services Limited & Urja Developers Private Limited	Estimated ER Sheet, Actual ER Sheet	Refer to Attachment
3	EKI Energy Services Limited & Urja Developers Private Limited	IRR Sheets	Refer to Attachment
4	EKI Energy Services Limited & Urja Developers Private Limited	Shareholding book	Refer to Attachment
5	EKI Energy Services Limited & Urja Developers Private Limited	Commissioning Certificates	Refer to Attachment
6	EKI Energy Services Limited & Urja Developers Private Limited	Authorization Letters	Refer to Attachment
7	EKI Energy Services Limited & Urja Developers Private Limited	Emission Reductions Purchase Agreement	Refer to Attachment
8	EKI Energy Services Limited & Urja Developers Private Limited	Power Purchase Agreements	Refer to Attachment

9	EKI Energy Services Limited & Urja Developers Private Limited	Supplier's agreement for equipment/instruments	Refer to Attachment
10	EKI Energy Services Limited & Urja Developers Private Limited	Local Stakeholder Consultation: Minutes of meetings	Refer to Attachment
11	EKI Energy Services Limited & Urja Developers Private Limited	Single Line Diagrams	Refer to Attachment
12	EKI Energy Services Limited & Urja Developers Private Limited	Feasibility Reports	Refer to Attachment
13	EKI Energy Services Limited & Urja Developers Private Limited	Joint Metering Reports (MP: 01, 17-September-2021 to 31-December-2022)	Refer to Attachment
14	EKI Energy Services Limited & Urja Developers Private Limited	Logbooks (MP: 01, 17-September-2021 to 31-December-2022)	Refer to Attachment
15	EKI Energy Services Limited & Urja Developers Private Limited	Invoices for Electricity Import/Export (MP: 01, 17-September-2021 to 31-December-2022)	Refer to Attachment
16	EKI Energy Services Limited & Urja Developers Private Limited	Training schedules	Refer to Attachment